

Optical Integrated Sensing and Communication for 6G Through a Systematic Review of Architectures, Metrics, and Tradeoffs Across Optical Platforms

Fatih Dönmez, Ahmet Altuncu, and Mustafa Namdar

Abstract—Optical integrated sensing and communication (O-ISAC) reuses optical infrastructure, signals, or hardware to transmit data and observe a physical environment. Its implementations span guided fiber, optical wireless, and photonics-enabled millimeter-wave and terahertz links, whose different sensing paths complicate the interpretation of reported performance. This systematic review synthesizes 227 eligible reports representing 206 studies published from January 2020 through June 2026. We connect each architecture’s shared resource to the design variable, physical mechanism, and communication and sensing outcomes it affects, retaining source-specific measurement and validation conditions. The synthesis identifies distinct routes to joint performance: reference allocation can reduce payload opportunity; increased probe power can improve sensing while impairing data recovery; and interference suppression or sensing-guided steering can support communication without the same resource exchange. These mechanisms explain why separately reported peak rates and sensing errors cannot define a common performance frontier. Laboratory and prototype evidence is broad, but only 12 studies reach field or deployment settings, with six reporting outcomes in both functional domains at that level. That coverage alone does not establish concurrent joint operation. We derive application-specific tests of traffic continuity, motion, optical drift, and inference cost, using paired measurements and separate-function baselines to determine when a demonstrated subsystem benefit remains useful in a network.

Index Terms—6G, integrated sensing and communication, optical communication, optical sensing, resource allocation, systematic review.

I. INTRODUCTION

A. Background and Motivation

Optical communication infrastructure can support data transmission alongside physical measurements, such as vibration along a fiber or receiver position in a room [1], [2]. Optical integrated sensing and communication (O-ISAC) uses this

capability to combine the two functions through shared infrastructure, hardware, or signals [3], [4]. This extends integrated sensing and communication (ISAC) to optical and photonic systems [5]. ISAC is included among the usage scenarios in the International Mobile Telecommunications framework for 2030 and beyond [6]. The opportunity to reuse resources brings a joint design problem, since their operation must account for both communication and sensing requirements.

How these resources are shared depends on the signal path and the sensing task. In fiber and optical wireless systems, light interacts with the sensing medium or reaches the receiver or target [3], [7], [8]. In photonics-enabled wireless systems, optical components can generate the signal, while propagation to the target occurs in the millimeter-wave or terahertz band [9], [10]. The common design question is how the shared resource changes the signal available to each receiver, and therefore the communication and sensing performance. Section II-A develops these paths in one physical diagram.

Shared operation can introduce a tradeoff or allow one function to assist the other. In one fiber experiment, raising the sensing-probe power while holding communication launch power fixed improved the sensing signal-to-noise ratio but increased the communication bit error rate [1]. In a visible-light prototype, position estimates guided beamforming toward the receiver and improved the communication data rate at the tested locations [2].

These different responses make resource reuse an insufficient basis for selecting an architecture. The relevant question is which physical mechanism connects the two outcomes and whether it remains useful under the application’s power, geometry, traffic, and timing constraints. This survey follows that connection from architecture through performance and validation to the tests needed for further development.

B. Related Surveys

Existing reviews provide several perspectives on these design questions, through platform-specific analyses and broader views of O-ISAC. A tutorial on optical architectures and waveform allocation uses a common simulation setting to show communication error rate and ranging error [11]. A review of lighting, sensing, and communication integration provides performance tables for visible-light and hybrid systems [12]. A waveform-focused review examines photonic millimeter-wave and terahertz demonstrations and compares integrated waveforms through models and experiments [13]. Other focused reviews address optical transmission and estimation, or

This review received no specific financial or nonfinancial support. No funder or sponsor had any role in the design, conduct, analysis, interpretation, manuscript preparation, or decision to submit.

Fatih Dönmez is with the Department of Electrical and Electronics Engineering, Afyon Kocatepe University, 03200 Afyonkarahisar, Türkiye (e-mail fatih.donmez@usr.aku.edu.tr, ORCID 0000-0003-0553-4418).

Ahmet Altuncu is with the Department of Electrical and Electronics Engineering, Faculty of Technology, Afyon Kocatepe University, 03200 Afyonkarahisar, Türkiye (e-mail aaltuncu@aku.edu.tr, ORCID 0000-0002-3753-9515).

Mustafa Namdar is with the Department of Electrical and Electronics Engineering, Faculty of Engineering, Kütahya Dumlupınar University, 43100 Kütahya, Türkiye (e-mail mustafa.namdar@dpu.edu.tr, ORCID 0000-0002-3522-4608).

The corresponding author is Fatih Dönmez (e-mail fatih.donmez@usr.aku.edu.tr).

communication and sensing in fiber networks [14], [15], [16], [17].

Broader surveys connect these technical developments to larger research questions. An evolution-oriented survey traces ISAC across spectrum, network scale, and sensing modalities, including optical and RF–optical systems [5]. A broad O-ISAC review brings together optical platforms, experimental results, waveform design, and applications [4]. These two surveys also discuss performance measures and the need for further evaluation. Together, these works explain design choices, report numerical demonstrations, and identify research directions.

Table I distinguishes the physical coverage and review approach of representative works. Its platform columns identify where the sensing interaction takes place. The last two columns address how studies are selected and how results from different studies are judged comparable. This distinction matters because a controlled waveform comparison and a comparison between independently reported experiments answer different questions. The latter also requires agreement on what was measured and under which conditions.

A further need is to explain when a reported result can guide another design. The difficulty of using common performance measures across RF and optical systems is discussed in [5], while shared benchmarks are identified as a research need in [4]. These observations motivate comparison rules that connect shared design choices to both performance outcomes and specify the measurement conditions under which results can be compared.

C. Review Methodology

We conducted a systematic literature review, with reporting guided by PRISMA 2020 and PRISMA-S [18], [19]. Searches covered six bibliographic and publisher sources and were completed on 22 June 2026. The publication window extended from 1 January 2020 to that date, using the date when a report first became publicly available. Search strategies combined O-ISAC terminology with terms for joint sensing and communication in optical or photonic systems. Eligible reports were peer-reviewed journal articles or full conference papers describing communication and sensing within a shared optical or photonic system, with sufficient English-language technical content to assess at least one survey domain.

After duplicate removal, records underwent title and abstract screening followed by full-text eligibility assessment. Figure 1 summarizes the selection flow and distinguishes publications from underlying studies. Contextual literature was retained separately for background and field positioning. We extracted reported results with their metric definitions, units, measurement points, and operating conditions, and used structured narrative synthesis to examine recurring mechanisms and performance relations. Numerical comparisons retained the conditions under which each result was obtained. The search strategies, eligibility criteria, study-selection records, and supporting review materials are available in the OSF repository.¹

¹<https://osf.io/7f6wb/>

D. Scope and Contributions

This survey examines O-ISAC in guided optical systems, optical wireless links, and photonics-enabled wireless systems, including hybrid configurations. We analyze how shared infrastructure, signals, and design objectives connect communication and sensing, and how these connections shape joint performance.

- **A physical basis for selecting an O-ISAC design.** We distinguish platforms by their sensing paths and identify how hardware, signals, and allocated resources connect the two functions. This explains which integration mechanisms fit a given physical setting and which constraints must be considered when adapting them (Sections II and III).
- **A common basis for interpreting performance.** We relate waveform and receiver choices, together with resource allocation, to communication rate and reliability alongside sensing accuracy and resolution. We distinguish relationships demonstrated within a study from comparisons that require compatible metrics, measurement points, and operating conditions (Sections II and IV).
- **Evaluation priorities derived from the evidence.** We derive joint tests from the validation limits observed across optical platforms. These tests pair communication and sensing measurements with separate-function baselines under changes in traffic, propagation conditions, or hardware stability (Sections V and VII).

Section II introduces the physical and performance concepts used in the review. Sections III through VI examine platforms, joint design, validation, and applications. Section VII develops the research priorities, and Section VIII concludes the article.

II. TECHNICAL FOUNDATIONS OF O-ISAC

The physical interaction that enables sensing also determines how it can share resources with communication. This section follows the signal from that interaction to the receiver, then explains the optical constraints on sharing and the measures used to evaluate both functions.

A. Signal Paths and Sensing Observations

A communication receiver recovers information encoded by the transmitter. A sensing receiver estimates a property of the target or propagation medium from its effect on the received signal. The two functions can use the same received samples, different observations of one transmission, or separate signals using shared infrastructure. Figure 2 separates signal generation, physical interaction, and receiver processing. This separation locates both the sensing mechanism and the point at which each performance metric is measured.

The sensing observation depends on the physical task. Fiber vibration can change the phase or polarization of a transmitted or backscattered field. Reflective ranging uses a return whose delay depends on the target distance. Visible-light positioning can instead infer receiver location from the intensities of known illumination patterns. These paths lead to different estimators even when they share optical components [1], [2], [20], [21].

TABLE I
PHYSICAL COVERAGE AND REVIEW APPROACH OF REPRESENTATIVE RELATED REVIEWS AND TUTORIAL-STYLE SYNTHESSES

Ref. / year	Main focus	Optical platforms			Review approach	
		Fiber	Optical wireless	Photonic wireless	Study selection	Cross-study comparison rules
[11] (2024)	Optical architectures, waveform design, and resource allocation	—	●	—	—	—
[12] (2024)	Lighting–sensing–communication integration and hybrid links	—	●	—	—	—
[13] (2026)	Integrated photonic waveforms and controlled performance comparisons	—	—	●	—	—
[5] (2026)	ISAC evolution across spectrum, networks, and sensing modalities	●	●	●	—	●
[4] (2026)	Optical platforms, experimental demonstrations, and applications	●	●	●	—	●
This survey	Shared-resource performance relations and evidence-based evaluation	●	●	●	●	●

●: structured treatment; ○: selected discussion or an identified research need; —: not developed in the stated sense. Fiber denotes sensing in the fiber, not fiber transport alone; photonic wireless denotes optically generated or processed signals with wireless RF propagation. Only optical-related coverage is marked for [5]. Study selection denotes an explicit search and eligibility procedure. Cross-study comparison rules address metric definitions, measurement points, conditions, baselines, and supporting evidence; they are distinct from a controlled comparison within one model or experiment.

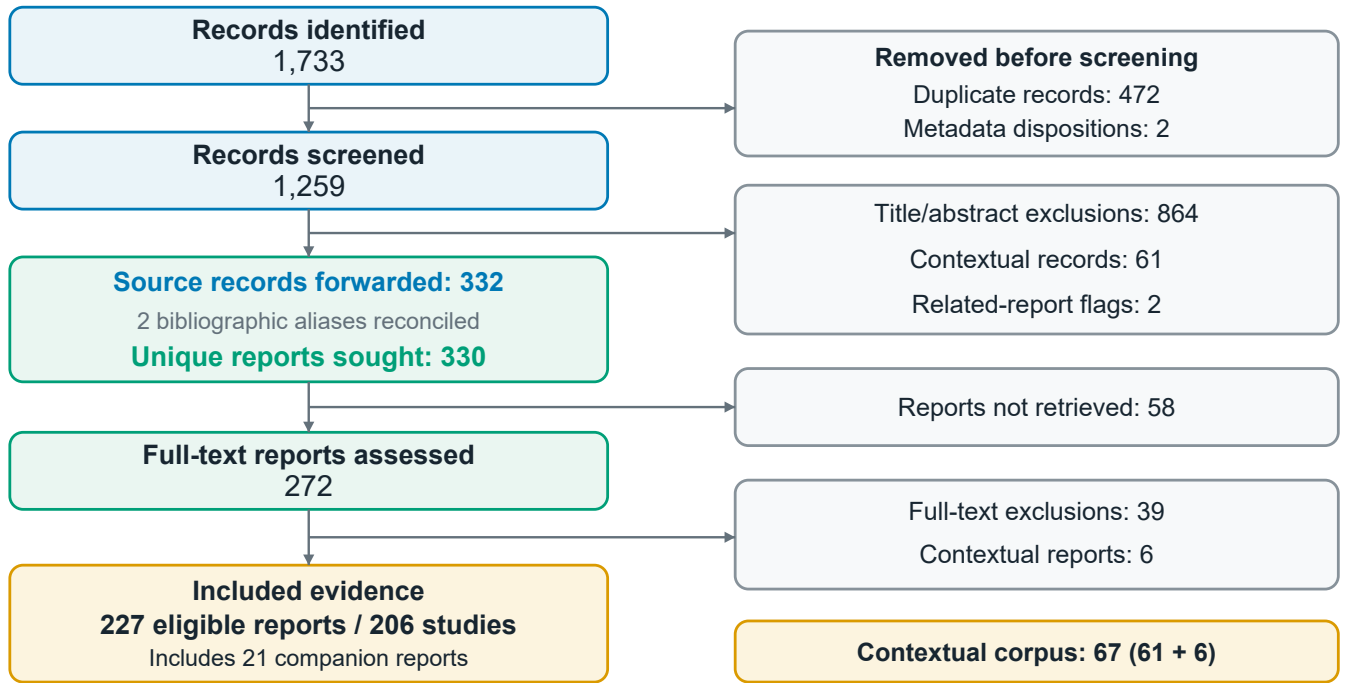


Fig. 1. Study selection for the O-ISAC review. The flow distinguishes search records, eligible reports, and underlying studies. The 227 eligible reports describe 206 studies; 21 companion reports remain linked to those studies. The 67 contextual records support background and field positioning. Detailed eligibility decisions are provided in Electronic Supplement S-Exclusions.

Receiver design determines which signal properties are available for estimation. For an intensity-modulated, direct-detection link operating in a linear intensity-channel regime, the electrical observation can be written as

$$y(t) = \mathcal{R}_{\text{pd}} [h_{\theta} * x](t) + n(t), \quad x(t) \geq 0, \quad (1)$$

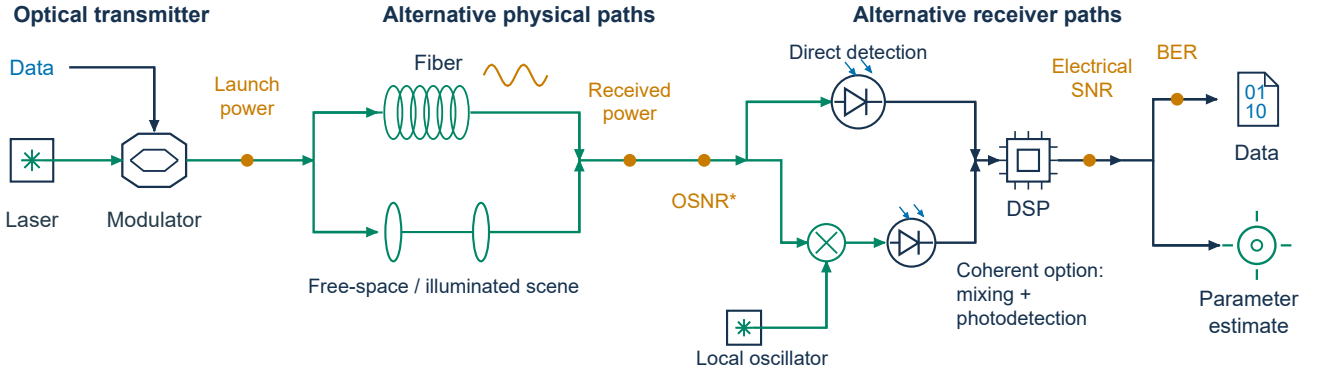
where t denotes time, $y(t)$ is received current in amperes, and $x(t)$ is transmitted optical power in watts. Photodetector responsivity $\mathcal{R}_{\text{pd}}$ converts optical power to current and has

units of amperes per watt. The intensity impulse response h_{θ} depends on the physical parameters θ to be sensed, and $*$ denotes convolution. The additive term $n(t)$ represents electrical noise at the detector output.

This model relates optical power to a current observation. Coherent reception mixes the received field with a local oscillator and provides complex observations for amplitude and phase processing [1], [11].

The role of optics changes in photonics-enabled wireless

(a) Optical propagation and observation



Observations may use forward light, backscatter, reflections, or spatial intensity. *OSNR where applicable.

(b) Photonic generation followed by RF propagation

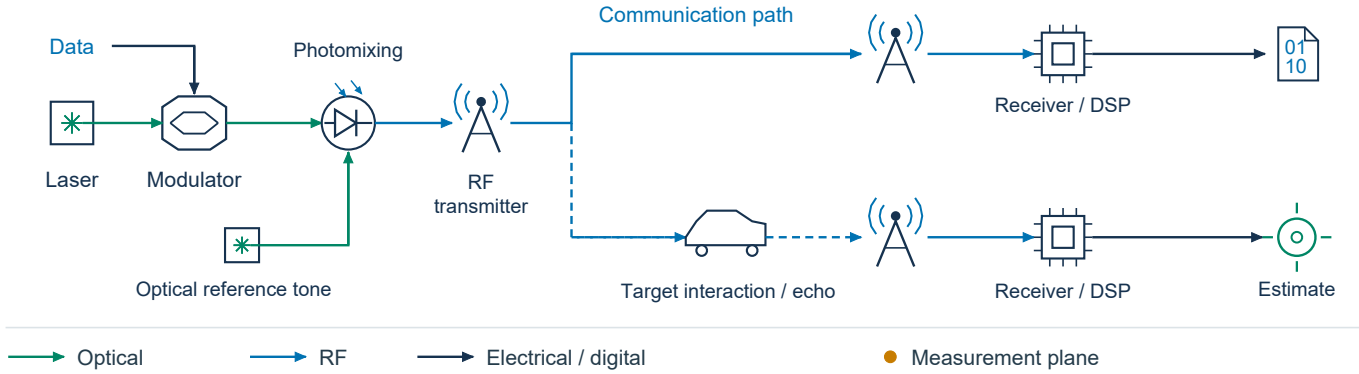


Fig. 2. Conceptual signal paths and measurement locations in O-ISAC. Panel (a) shows alternative fiber and free-space optical paths with direct or coherent reception. Sensing observations may use forward light, backscatter, reflections, or spatial intensity. Panel (b) separates optical generation from radio-frequency (RF) propagation and distinguishes communication and target-interaction paths. Receiver icons represent functional roles with shared or separate hardware. Panel (a) illustrates a laser and external modulator; an intensity-driven LED can supply the optical waveform in an IM/DD implementation. Markers identify launch and received optical power, optical signal-to-noise ratio (OSNR) before photodetection where applicable, electrical signal-to-noise ratio (SNR) at the stated electrical receiver plane, and bit error rate (BER) at data recovery.

systems. Optical tones can be mixed to generate a millimeter-wave or terahertz carrier at their difference frequency. The target is then interrogated by a radio-frequency (RF) signal, although optical sources and modulators determine its waveform. For example, a 262.1–268.3 GHz transmission is generated by mixing a swept optical source with a fixed optical tone [10]. Optical-source stability and wireless propagation therefore enter different stages of the same sensing and communication chain. Locating the interaction and receiver first makes it possible to identify the resources that each function can reuse.

B. Shared Resources and Optical Signal Constraints

Along the signal path, the shared resource may be a component, a transmission interval, or the waveform itself. Time sharing assigns communication and sensing to different slots. Frequency sharing assigns separate spectral portions, often with a guard band to limit interference. A joint waveform carries data while retaining structure that supports parameter estimation [11], [13]. Figure 3 illustrates these allocation

patterns. Shared infrastructure can support any of them. The allocation determines which time and frequency resources are available to each function, while hardware constraints determine the waveform that can occupy them.

Optical hardware constrains the waveform that can realize this sharing. In an intensity-modulated link, a bipolar electrical waveform must produce nonnegative optical power within the source's usable range. A clipped model with a direct-current (DC) bias makes this constraint explicit,

$$x(t) = \min\{P_{\text{pk}}, \max[0, b + a u(t)]\}, \quad (2)$$

$$\mathbb{E}[x(t)] \leq P_{\text{av}},$$

where $u(t)$ is a dimensionless real waveform, while a and b specify the modulation amplitude and bias in optical-power units. The peak and average power limits are P_{pk} and P_{av} , with $\mathbb{E}[x(t)]$ denoting mean optical power. Increasing the bias reduces clipping at zero but consumes average power and leaves less headroom below the peak limit. Clipping distorts the waveform used for data recovery and parameter estimation, so bias and modulation amplitude must be chosen together [11].

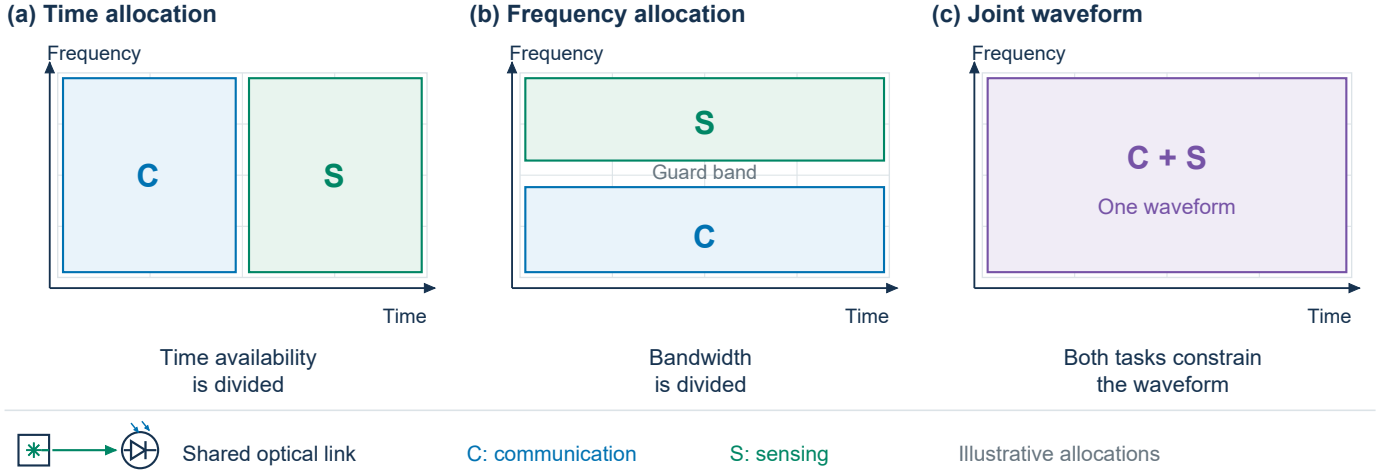


Fig. 3. Conceptual time-frequency allocations for communication and sensing. Panel (a) assigns different time slots to the two functions. Panel (b) assigns separate frequency bands, with a guard band shown in this example. Panel (c) uses one waveform within a shared allocation. The horizontal and vertical axes denote time and frequency; C and S denote communication and sensing. Source and detector icons indicate a shared optical link, while rectangles indicate allocated resources. Equal panel dimensions serve the illustration and do not assert equal bandwidth or performance.

The power constraint applies to optical intensity. Complex coherent samples and the bipolar RF waveform after photonic conversion have different signal representations. In particular, nonnegative transmitted optical power does not require each electrical sample to be nonnegative.

An optical orthogonal frequency-division multiplexing (OFDM) simulation illustrates the bias choice. It uses 256 subcarriers spaced by 3.9 MHz and 32-symbol frames in a 200 m free-space optical channel model [11]. The nominal subcarrier grid spans $256 \times 3.9 = 998.4$ MHz, approximately 1 GHz. In this model, insufficient bias increases communication bit error rate through clipping, while excessive bias assigns more power to a component that carries no data. Range estimation is more tolerant of the clipping in the simulated settings, so its root mean square error responds differently to the same bias choice.

Sharing can also connect the functions through receiver information or spatial waveform structure. In a fiber experiment, a frequency-response estimate from the communication receiver is used to pre-compensate a sensing probe [1]. In a visible-light system, averaging the received pattern response supplies spatial information, while high-pass filtering recovers the temporal data modulation [2]. These mechanisms explain how the functions interact. Evaluating their effect requires measures that retain the meaning of each output.

C. Communication and Sensing Measures

The effect of a shared setting depends on what each performance measure counts. Communication rate describes how information is counted over time. For a frame carrying N_{payload} useful bits over a duration T_{frame} in seconds, net rate in bits per second is $R_{\text{net}} = N_{\text{payload}}/T_{\text{frame}}$. Symbol rate instead counts transmitted symbols, and its conversion to bit rate depends on modulation and overhead. Bit error rate (BER) measures the fraction of recovered bits that are incorrect at the stated processing stage. Delivered throughput also depends on decoding success and retransmissions. A rate estimated from

received signal quality describes a supported transmission rate under the assumed modulation and coding [2], [22], [23].

Sensing metrics distinguish target separation from parameter estimation. For two-way ranging in air, delay τ determines range r , while an ideal swept waveform gives the familiar nominal resolution

$$r = \frac{c\tau}{2}, \quad \Delta r \simeq \frac{c}{2B_s}, \quad (3)$$

where c is the propagation speed in air and B_s is the sensing sweep bandwidth in hertz. Delay is measured in seconds and range in metres. The resolution expression assumes a calibrated, bandwidth-limited delay response. Windowing, processing, and target conditions influence practical separation. A fiber delay model requires the appropriate propagation speed in the medium.

Estimation error answers how closely an estimated parameter matches its reference value. For N range estimates, the root mean square error (RMSE) is

$$\text{RMSE}_r = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{r}_i - r_i)^2}. \quad (4)$$

Here, $\hat{r}_i$ is the estimated range and r_i is its reference value for observation i , with both expressed in the same units. A maximum absolute error describes a different statistic. For example, a 6.2 GHz sweep gives a nominal resolution of about 2.4 cm, whereas the experiment reports a maximum ranging error of 0.55 cm for static targets at 20–45 cm [10]. This smaller error is consistent with estimating one target more precisely than the nominal separation scale for two targets. Practical resolution is established through a two-target measurement. A theoretical estimation bound instead describes the limit implied by its observation and noise model, which must be specified when comparing it with measured error [11].

The sensing task also determines the meaning of a spatial dimension. Fiber spatial resolution describes localization along

the cable, while gauge length specifies the separation over which differential phase is evaluated. The fiber example in Table II reports both quantities for this reason [1]. Visible-light localization likewise separates lateral and depth performance. Its ability to distinguish displacements along either direction is separate from the rate at which a moving receiver's location is updated [2].

Signal quality must likewise be tied to a point in the receiver chain. The optical signal-to-noise ratio (OSNR) compares optical signal and noise power in a stated reference bandwidth before photodetection. Electrical signal-to-noise ratio (SNR) depends on the detector, noise model, and electrical processing bandwidth. Launch power and received power similarly refer to different locations in Fig. 2. These distinctions explain why equal numbers in decibels need not represent equal receiver conditions [20], [23]. Once these meanings and measurement points are fixed, communication and sensing outcomes can be related to the setting that produced them.

D. Joint Operating Points and Performance Interpretation

A joint operating point links a shared design setting to both communication and sensing outcomes under a common, identified condition set. The results may come from one acquisition or from tests that explicitly retain the conditions needed for the claimed relationship. A claim of simultaneous operation additionally requires evidence that both functions operate together. Shared hardware alone establishes where integration occurs, while the performance relation depends on this connection between outcomes and settings.

Table II illustrates the meanings of reported measures and their evaluation conditions in three systems. Its rows collect results from the respective studies, including separate tests, and should be read as metric examples rather than single jointly measured operating points. The distinction preserves the contribution of each experiment while identifying which values can be interpreted together.

A shared waveform can support both tasks while their reported numerical results come from different receiver configurations or tests, as in the photonic-THz and visible-light examples. A controlled sweep instead links a specific change to both responses. Section IV examines such within-study relationships, including the fiber probe-power sweep, and distinguishes them from benefits obtained by sharing receiver information.

Once both outcomes can be evaluated for a shared setting, joint design can express the choice as a constrained problem. An illustrative sensing-oriented formulation is

$$\begin{aligned} \min_{\mathbf{u} \in \mathcal{U}} \quad & \text{RMSE}_s(\mathbf{u}) \\ \text{subject to} \quad & R_{\text{net}}(\mathbf{u}) \geq R_0, \\ & \text{BER}(\mathbf{u}) \leq \epsilon_b, \end{aligned} \quad (5)$$

where RMSE_s is the root mean square error in the selected sensing quantity. The vector $\mathbf{u}$ contains the shared design variables, and $\mathcal{U}$ imposes power, bandwidth, and hardware constraints. The service requirements R_0 and ϵ_b fix the communication performance that must be retained. This formulation illustrates a service constraint, without assuming that

every cited study solves this problem. A communication-oriented formulation instead maximizes a communication objective subject to a sensing requirement. A Pareto analysis examines settings for which improving one selected objective requires worsening the other within the feasible set [11], [13]. Establishing that frontier requires evaluating the feasible alternatives. A collection of separately reported best values does not establish it.

Interpreting results across studies adds physical and measurement conditions to this design view. Numerical comparisons require independent studies with compatible tasks, metric definitions, and measurement points. Their operating conditions and baselines must support the particular comparison, and a joint claim requires both outcomes to be linked to the shared setting. For an integration-gain comparison, a separate-function baseline must preserve the relevant resource budget and task conditions. Otherwise, a change in available power or measurement time can be mistaken for a gain due to sharing. When magnitudes are not directly comparable, a recurring mechanism can still explain a design choice, with each result retained in its own context.

The architecture analysis in Section III now identifies which shared choices each physical platform makes available. Their measured or modeled consequences follow in Section IV.

III. OPTICAL PLATFORMS AND ARCHITECTURE CHOICES

The signal paths in Section II leave an important design question: where should communication and sensing share a component, signal, or decision? Reusing a communication carrier for sensing exposes the measurement to its modulation and reference errors. Adding a sensing probe provides more control over the measurement but introduces an additional signal into the link. Moving the target interaction beyond an optical-to-radio conversion adds another channel and calibration boundary. The appropriate architecture therefore follows from the location of the target and the resources through which the two functions interact.

Table III connects the platform families to these decisions. The optical wireless families include visible light communication (VLC), light fidelity (LiFi) networking, and free-space optical (FSO) links. Each study has one dominant platform assignment; a system can share several resources and contain more than one propagation medium. The discussion follows the physical path, then examines how its shared elements become design variables.

A. Guided Paths: Sensing the Fiber or Transporting the Probe

When fiber is the sensing medium, environmental disturbances alter the same guided field that carries traffic. Backscatter, carrier phase, modal coupling, or an inserted sensor can reveal distributed events, vibration, strain, and environmental change [24], [25], [26], [27]. The architecture first decides whether to recover this information from the communication signal or to introduce a dedicated sensing signal. Carrier and waveform reuse make symbols, phase, or supported modes part of the measurement [28], [29], [30]. Separate probes or

TABLE II
THREE EXAMPLES OF METRIC INTERPRETATION UNDER SOURCE-SPECIFIC TEST CONDITIONS

Configuration	Reported communication and sensing measures	Interpretation
Fiber [1]: 40 km; two 30 GBaud 16-QAM subcarriers; 500 MHz LFM probe; 200 Hz vibration on a 1 m segment.	2×120 Gbit/s during concurrent sensing; nominal spatial resolution 1 m with a 10 m gauge length. Probe pre-compensation improves vibration-spectrum SNR by 2.4 dB.	Gauge length differs from spatial resolution. The 2.4 dB gain is a pre-compensation result, separate from the probe-power sweep, which fixes communication launch power at 0 dBm rather than total power.
Visible light [2]: 225 mW white laser; 1 GHz receiver; 0.6×0.6 m illumination area at 2 m.	Estimated rate averages 3.35 Gbit/s (3.16–3.52 across patterns). Separate tests resolve 1 mm lateral and approximately 4 cm depth displacements; dynamic localization runs at 39 frames/s.	Estimated rate is not delivered throughput. Pattern averages, static spatial tests and dynamic localization describe different evaluations.
Photonic THz [10]: 262.1–268.3 GHz ASK/FMCW signal; 6.2 GHz sweep in 400 ns; communication distance 1 m.	15 Gbit/s with BER 2.97×10^{-3} ; approximately 36.2 GHz occupied bandwidth. Nominal range resolution 2.4 cm; maximum error 0.55 cm for static targets at 20–45 cm in a separate receiver configuration.	Sweep and occupied bandwidth differ. Range error is distinct from resolution; target range is distinct from communication distance.

Values in a row may come from separate tests and do not by themselves define one jointly measured operating point. QAM: quadrature amplitude modulation; LFM: linear frequency modulation; ASK: amplitude-shift keying; FMCW: frequency-modulated continuous wave.

TABLE III
PHYSICAL PATHS AND ARCHITECTURE CHOICES ACROSS THE 206 STUDIES

Platform; studies, n (%)	Target interaction and observation	Main shared choices	Conditions that constrain the choice
Photonic terahertz; 69 (33.5%)	Optical generation followed by a wireless millimeter wave or terahertz target path	Reference, conversion, waveform and beam control	Source coherence, conversion bandwidth, antenna geometry and wireless propagation
Fiber; 56 (27.2%)	Guided-path perturbation, or a downstream target reached through optical transport	Communication-carrier reuse, separate probe, wavelength and receiver allocation	Leakage, nonlinear interaction, topology, traffic and remote calibration
VLC and LiFi; 38 (18.4%)	Direct and reflected light in an illuminated scene	Modulation, exposure, illumination and spatial allocation	Source response, ambient light, field of view, orientation and receiver type
Free space optical; 31 (15.0%)	Directional optical link and, where present, target return	Waveform, aperture, beam and acquisition time	Pointing, divergence, collected power, reflectivity, weather and motion
Hybrid optical; 9 (4.4%)	Interaction across optical, guided and wireless segments	Shared reference, conversion or sensing-guided control	Segment power, synchronization, calibration and control timing
Other optical; 3 (1.5%)	Microcomb-generated microwave field, ultraviolet response or quantum phase	Specialized source, reference and task design	The native propagation, receiver and measurement model
Total; 206 (100.0%)	One dominant family per study; physical segments and sharing mechanisms may overlap.		

wavelengths permit a different interrogation structure while retaining common fiber and receiver constraints [3], [31], [32].

These alternatives place interference at different points. A separate probe can leak into the data channel or induce non-linear interaction, so its launch conditions affect both sensing support and communication margin. Reusing the data carrier avoids that particular addition but makes sensing depend on the available reference and on how data modulation is removed. Coherent vibration sensing, for example, relies on preserving and recovering phase along the communication path [28]. Dynamic range, reflections, topology, and traffic determine which reuse is practical; their importance persists in live access networks and deployed routes [24], [32].

Fiber used to transport a probe has a different measurement boundary. The optical link distributes signals to a radio frequency identification (RFID), radio, or radar segment, where the target interaction occurs [33], [34], [35], [36], [37]. The recovered quantity may be tag state, range, localization, or an image. Optical transport then contributes conversion, timing, and reference errors to an observation also shaped by wireless geometry. A reported transport span therefore cannot substitute for target range. Evaluating this architecture requires following

the outgoing signal and the return through their conversion and clocking stages.

B. Illuminated Scenes and Directional Optical Links

In VLC and LiFi systems, the emitted light simultaneously defines the communication path and the scene available for observation. Photodiodes favor rapid temporal reception, while cameras provide spatially resolved samples subject to exposure and readout constraints. Direct and reflected paths can support position, pose, gesture, recognition, and physiological sensing [38], [39], [40]. The receiver choice thus changes what can be inferred from the common optical field, as well as which modulation and update rates are usable.

Illumination makes waveform and spatial design interdependent. Nonnegativity, dimming, flicker, and source linearity restrict temporal modulation, while ambient light, orientation, and field of view determine which signal reaches the detector [41], [42]. A design combining temporal data with structured illumination uses position estimates to adapt beamforming; the resulting coupling runs from the observed geometry back to the transmitted field [2]. Other designs allocate emitters or spatial resources to communication and localization [43], [44],

[45]. Here, a stronger received signal and a more informative localization geometry are distinct objectives. Their relation depends on emitter placement and receiver orientation.

FSO links place similar spatial decisions in a directional beam and aperture path. The link may deliver data directly, illuminate a target, or collect a retroreflected return. Aperture, divergence, pointing, reflectivity, and motion then determine the available communication power and sensing observation [46], [47], [48]. Scanning and steering choose which region is observed and served [49], [50], [51]; atmospheric estimates can support beam shaping or link selection [8], [52]. A narrow spatial response can aid selection of a user or target, but its value depends on maintaining alignment. Thus, the aperture and control strategy belong to the performance result, together with the waveform. Laboratory ranging and sustained operation under changing atmospheric conditions test different parts of this dependency [8], [48].

C. Photonic Generation with Wireless Target Interaction

Photonic millimeter wave and terahertz systems use optics to generate, distribute, or process a carrier before a wireless target interaction. Laser coherence and linewidth affect the shared reference; component bandwidth and conversion fidelity determine the signal delivered to the antenna [53], [54]. Antenna aperture, alignment, molecular absorption, and target scattering subsequently shape the radiated link and return [55], [56]. This separation identifies which impairment a proposed photonic improvement can address and which remains in the wireless path.

Architecture determines how far the common reference extends. A multicarrier frame can allocate spectral support between data and sensing, while a photonically controlled array places the joint decision in the transmitted beam [9], [22]. Wavelength tuning or multibeam reception moves control to the source or receiving front end [57], [58]. Combining linear frequency modulation (LFM) with communication symbols across coherent carriers makes waveform generation a shared constraint [53]. Frequency hopping and delay-Doppler multiplexing provide other ways of structuring that signal [56], [59]; their benefits depend on how the receiver recovers the communication symbols and target response.

The return path can introduce a further optical stage. An echo may be remodulated onto light for centralized processing, with photonic dechirping mapping delay to a lower-frequency observable [60], [61]. Reference phase noise degrades recovery, and optical/electrical path mismatch can shift the inferred range [60]. A shared source consequently reduces neither all timing errors nor all calibration requirements. Controlled experiments expose these conversion and beamforming effects [9], [53]; the reported 270 m outdoor link also tests operation across the complete wireless chain [22].

D. Integration Across Segments and Control Loops

Hybrid integration becomes consequential when conversion or feedback between segments changes the system response. Fiber and wireless interactions can couple power, timing, and calibration across a common waveform [62], [63]. A shared

photonic source can span fiber, FSO, and millimeter wave stages [64]. In each case, an end-to-end result depends on the interfaces as well as on the individual segments. Separate segment measurements establish only the behavior actually measured at those boundaries.

A control loop creates another form of integration. Sensing in one segment may guide service in another, as in blockage-aware access and combined optical delivery and monitoring [65], [66]. Sensed FSO channel state can inform trajectory and resource allocation, or access placement and user association across optical backhaul and radio access [67], [68]. The shared element is the decision: observation age, processing, and actuation now matter alongside signal quality. Section VI examines the service requirements placed on these loops.

The remaining specialized paths retain their native measurement models. A microcomb-generated microwave field, a solar-blind ultraviolet response, and a quantum phase measurement provide different observations and receiver requirements [69], [70], [71]. Likewise, integration of communication and physiological sensing in a common material platform can retain separate detector roles [72]. Such examples extend the architecture space without establishing equivalent communication or sensing tasks.

E. From a Shared Component to a Joint Design Choice

Across these paths, the decisive step is to identify an adjustable choice. Hardware sharing makes a component's bandwidth, coherence, or dynamic range common to both functions. Carrier sharing supplies a frequency or phase relation. Waveform sharing goes further by making a pilot, chirp, code, or spatial pattern contribute to decoding and sensing. In OFDM, the chosen pilots and payload structure determine which samples are known to the receiver and how much signal support remains for data [73]. In guided sensing, removing data modulation to recover phase makes receiver processing part of the integration [28].

The resulting evidence depends on how the shared choice was exercised. One photonic prototype uses frequency steps for the radar dimension and amplitude for data, with its experimental support strongest for waveform generation and communication response [74]. An exposure-time design provides an analytical and numerical coexistence result, while laboratory communication and sensing gains were obtained in separate configurations [41]. Another photonic system uses a shared ASK/FMCW waveform, with functional metrics evaluated in separate receiver configurations [10]. Localization and imaging performed before data delivery on a shared source constitute a further temporal arrangement [49]. These distinctions determine which paired outcomes can be assigned to one operating point. The next section follows the shared choice through its mechanism to those outcomes.

IV. SHARED DESIGN CHOICES AND JOINT PERFORMANCE

A shared architecture becomes a performance question when a design variable changes. Increasing pilot support can improve a reference while reducing the data-bearing fraction;

increasing probe power can strengthen sensing while introducing interference; changing beam direction can alter both collected power and the geometry available for estimation. The useful synthesis follows the adjustable variable, the physical or processing mechanism, and the two responses under the reported conditions. It also distinguishes a measured response to a sweep from a modeled optimum or a proposed design relationship.

Figure 4 summarizes this reasoning. The reviewed evidence contains 402 substantive relationships from 168 studies, including quantitative and qualitative descriptions. Their prevalence is reported in Electronic Supplement S-Core; here, the emphasis is on how the mechanisms work. Table IV provides selected within-study anchors. Its rows illustrate different mechanisms and evidence forms rather than a common performance ranking.

A. Allocating Signal Space and Observation Time

When communication and sensing occupy a finite frame, an allocation changes both the samples available to the estimator and the opportunity to carry payload. Embedded-pilot OFDM makes this connection explicit. Increasing pilot support improves the reported pilot-deviation proxy, while BER and the data subcarrier fraction follow the selected allocation [73]. That proxy describes errors in known pilot symbols in a simulated channel; it does not establish target range or localization accuracy. The study's spectral-efficiency definition counts data subcarriers and treats cyclic-prefix overhead separately. Thus, the same allocation informs three reported outcomes, but each retains a different meaning.

Reserved intervals, subbands, and chirps extend the mechanism to other waveforms. Protection can isolate a sensing return, and a chirp can supply pulse-compression support, at the cost of occupying a particular part of the available signal space [31], [63]. In a photonic terahertz prototype, net communication rates depend on subcarrier assignment, payload occupancy, and coding overhead [22]. The resource cost therefore follows from the implemented frame, including the information needed for synchronization and recovery. It cannot be inferred from the modulation order or nominal bandwidth alone.

Full waveform reuse changes this accounting: sensing can use samples that also carry data, but the receiver must recover a reference or separate the two signal components. Full-grid OFDM reuse and shared optical waveforms illustrate this alternative [79], [80]. A wider usable bandwidth can support communication signaling and finer delay discrimination at the same time. The benefit is limited by the device response, reference quality, sidelobes, and receiver processing [53], [61], [81]. A bandwidth ratio is therefore informative only with the total bandwidth, target range, and receiver architecture [82].

Observation time creates a related, but distinct, allocation. Camera exposure, acquisition duration, and task scheduling affect sensing support and update cadence alongside data recovery or execution time [41], [83], [84]. Longer observation does not merely consume a frame fraction: it can change which temporal variations remain observable. Chirp repetition,

coding traffic, and embedded pulse count likewise place the two functions in a common schedule [39], [85], [86]. The exposure-time coexistence analysis in Table IV is especially instructive because the modeled shared choice and the separately obtained laboratory gains support different conclusions [41].

B. Distributing Power Under Interference and Receiver Limits

Power enters the joint design through a reference plane and a constraint. Changing a probe-to-data ratio at fixed communication launch power adds a different burden from dividing a fixed total budget. Launch power, received power, carrier-to-signal ratio, modulation swing, and detector bias refer to different points in the chain. Their effects pass through loss, noise, nonlinear interaction, and saturation before either functional metric is formed. Consequently, a reported gain can be interpreted only after identifying what power was varied and what remained fixed.

The fiber probe sweep in Table IV exhibits a direct conflict. Raising the sensing-to-telecom ratio at fixed communication launch power improves sensing SNR and raises BER, with the strongest communication penalty near the probe [1]. Related fiber designs connect probe or carrier ratios to phase, polarization, and vibration observability under their own OSNR and receiver conditions [3], [20], [87]. These observations locate the interaction in the optical link; they do not justify transferring a preferred ratio between different fiber topologies or receivers.

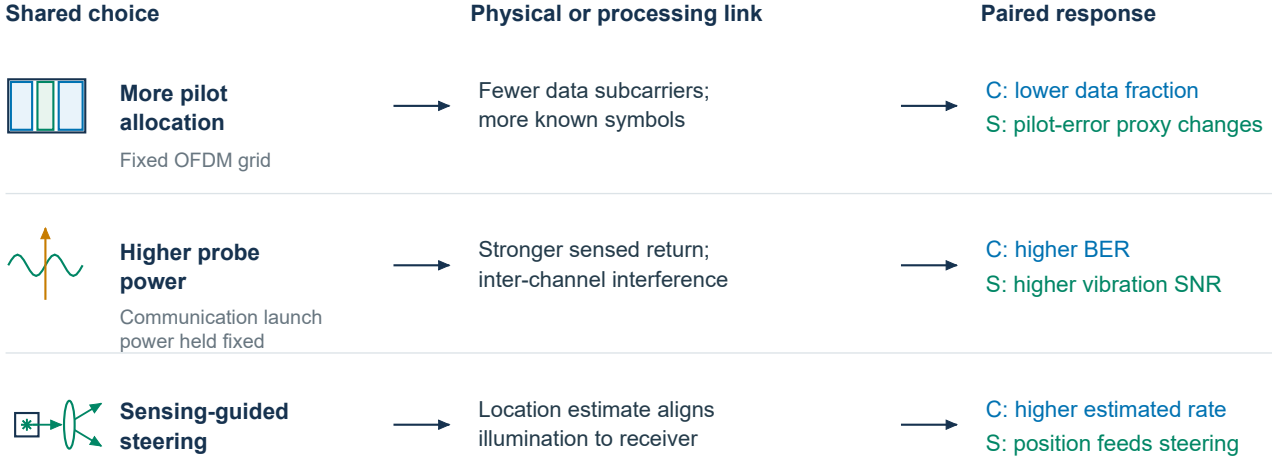
Architecture determines where that power is spent. Polarization paths, receiver taps, and separate service branches distribute the signal before estimation [88], [89], [90]. In a visible-light prototype, allocation between a communication/ranging branch and an energy-harvesting branch governs their reported outcomes under distinct receiver and endpoint bias conditions [91]. In photonic wireless systems, carrier and waveform ratios interact with phase recovery and the headroom available for communication and radar reception [60], [92], [93]. The same allocation can therefore produce different responses after a change in receiver architecture.

Absolute-power and drive sweeps expose another limit. Fiber launch conditions affect nonlinear interaction and sensing reach [23], [94], [95]; photonic front ends add modulator, converter, and amplifier limits [34], [96], [97]. Once a component limits recovery, additional signal power need not provide proportional improvement. Loss, pointing, and branch insertion loss also determine the margin actually available at the detector [32], [50], [98]. The useful operating region is bounded by these physical limits and by the communication and sensing thresholds required together.

C. Choosing Geometry, Aperture, and Spatial Support

Spatial design changes both received energy and the information carried by the available paths. Directing more light toward a communication receiver can increase its signal, while sensing also depends on the positions and angles needed to distinguish a target or estimate a location. Beam steering,

Two allocation costs and one feedback benefit



Directions apply to the reported configurations, not pooled effects. The pilot-error proxy is not target error. Position feeds steering; separate localization tests do not establish a joint accuracy gain.

Fig. 4. Two allocation costs and one feedback benefit in reported configurations. More pilots occupy data subcarriers in a fixed OFDM grid; the associated sensing measure is a pilot-error proxy [73]. Higher probe power at fixed communication launch power increases sensing SNR and BER [1]. A position estimate used to steer illumination supports higher estimated communication rate [2]; this does not imply a simultaneous improvement in localization accuracy. Arrows explain these within-study mechanisms and do not represent pooled trends.

access point placement, and spatial allocation therefore influence coverage and observability through related but unequal mechanisms [9], [99], [100]. A common physical path does not by itself imply that the communication receiver and sensing target favor the same beam.

Joint optimization expresses this dependence through an explicit feasible region. In the optical phased array simulation in Table IV, precoding and photodiode orientation are selected to maximize sensing contrast while satisfying a communication quality threshold and emitter power constraints [77]. The result establishes performance under the modeled scene and constraints. A related spatial design shows communication gain and target-profile estimation benefiting under alignment conditions [51]. Such joint gains are plausible when the useful spatial support overlaps; the stated geometry is part of the explanation, rather than an incidental test detail.

Geometry also enters through feedback. Structured illumination can provide a position estimate used to adapt communication beamforming [2]. Blockage-aware access can select a different usable path [65]. The control decision may improve the subsequent link, but its benefit depends on whether the observation remains current when the action occurs. This creates a bridge from static spatial optimization to the service timing examined in Section VI.

Spatial claims must also retain the implemented receiver configuration. Source directivity and tag-delay spacing affect coverage and the separability of returns [101], [102]. In another prototype, four surrounding receivers were designed but one was operated [103]. The measured configuration supports conclusions about that receiver path; array-wide coverage and crosstalk require their own tests.

D. Designing Waveforms and Recovering Shared Information

Waveform design can change the origin of interference as well as the allocation of signal energy. This distinction explains why superficially similar metric pairs exhibit different trends. In fiber communication with coherent reflectometry, shaping sequence transitions reduces nonlinear interference, lowering the communication penalty while retaining phase sensing at the reported condition [75]. A millimeter wave design instead increases a DC offset to strengthen the sensing component: radar signal-to-interference ratio improves while communication EVM rises [76]. The former modifies the interference mechanism; the latter changes the balance of energy within the waveform.

Sequence timing governs another set of coupled outcomes. Code length and period can connect signaling rate with delay resolution and unambiguous range [104], [105]. Chip duration, amplitude-keying timing, and guard duration likewise determine the signal support for data recovery and range discrimination [106], [107], [108]. Embedding communication data into a sensing signal can change its correlation or distributed-sensing fidelity [28], [109]. Reported rate-range relations therefore require the precise timing and ambiguity definitions; resolution, estimation error, and maximum unambiguous range remain different outcomes.

Receiver processing can also reuse information obtained by the other function. In the fiber probe study, a communication-receiver estimate of the link frequency response is used to pre-compensate the sensing probe, improving vibration-spectrum SNR by 2.4 dB in that processing test [1]. This result concerns compensation using shared receiver information; it is separate from the probe-power sweep and its communication penalty.

Receiver processing also decides how the acquired samples

TABLE IV
SELECTED MECHANISMS AND THEIR WITHIN-STUDY PERFORMANCE EVIDENCE

Shared choice	Communication response	Sensing response	Interpretation conditions and evidence
Embedded pilot allocation [73]	Changes BER and the data subcarrier fraction	More pilot support improves the reported pilot-deviation proxy	Simulation; known-pilot error is the reported sensing measure. The data fraction and cyclic-prefix overhead are defined separately.
Probe-to-telecom power ratio [1]	Higher ratio raises BER; the strongest penalty occurs near the probe	Higher ratio improves sensing SNR	Fiber probe sweep at fixed communication launch power; retain probe location and receiver conditions.
Sequence-transition shaping [75]	Reduces the communication penalty from nonlinear interference	Retains phase sensing at the reported condition	Shared fiber link with coherent reflectometry; the control changes how interference is generated.
Waveform direct-current (DC) offset [76]	Error vector magnitude (EVM) rises as the sensing component is strengthened	Radar signal-to-interference ratio rises	Millimeter wave waveform; energy redistribution changes the two recovery conditions. Higher EVM indicates worse communication signal quality.
Camera exposure time [41]	Exposure is part of the modeled communication/coexistence design	Exposure also governs the modeled sensing response	Analytical and numerical coexistence; laboratory gains for the two functions came from separate configurations.
Precoding and photodiode orientation [77]	Communication quality is constrained above a stated threshold	Sensing contrast is maximized within that feasible region	Optical phased array simulation; emitter power, scene, orientation, and metric definitions constrain the solution.
Source and carrier-recovery design [78]	Coherent data are recovered using broad-linewidth lasers	Vibration is recovered through residual-carrier processing	Long-span fiber experiment; hardware reuse moves requirements into phase recovery and digital processing, without a universal cost ranking.

become estimates. Spectral weighting and transform choices alter the recovered response and computational burden under a stated acquisition window [110], [111], [112]. Zero padding refines the sampled transform grid; it does not add independent delay information or, by itself, improve the physical separability set by the acquired signal. Similarly, changing a sensing estimator alone does not establish a joint communication-sensing improvement. Kalman tuning acts within the sensing branch [113], whereas loss weighting in a joint inference design explicitly connects communication reliability with distance and velocity errors [114]. The metric and branch affected by the processing change must therefore remain visible.

Source and digital processing choices can exchange implementation burden. One terahertz design uses coherent comb generation to stabilize quadrature phase-shift keying (QPSK) and LFM paths [61]. A long-span fiber experiment recovers coherent data and vibration using broad-linewidth lasers and residual-carrier processing [78]. These examples place reference recovery in different parts of the chain. They support architecture-specific feasibility, while a cost or energy comparison would require matched hardware, processing, and performance assumptions.

The processing burden also has a time dimension. Receiver bandwidth, converter rate, memory, and acquisition length determine how quickly a result is available. The outdoor terahertz study reports pipeline latency through hardware analysis and communication/ranging performance through a field experiment [22]. A smartphone camera system evaluates recognition and communication in the laboratory and module latency, memory, and complete-cycle timing on the device [84]. These measurements describe complementary parts of the implementation; their endpoints must align before they constitute an end-to-end latency result.

E. What the Reported Operating Points Establish

The mechanisms above explain both conflict and shared benefit. Allocating a finite resource can favor one task at the expense of another. Suppressing a common impairment or aligning useful spatial support can instead improve the conditions for both. A source-reported parameter sweep, a constrained model, and two separate demonstrations establish different parts of that argument. In particular, a modeled optimum remains conditional on its objective, constraints, noise model, and geometry; it is not a measured frontier for all systems using the same platform.

The comparison conditions in Section II determine whether the magnitude of a response can inform another design. Where those conditions differ, the transferable insight is the mechanism and the constraint that limits it. The remaining question is whether that explanation survives physical disturbances and can be tested independently. Section V examines the corresponding validation evidence.

V. JOINT VALIDATION UNDER REALISTIC CONDITIONS

The performance relations in Section IV identify how a shared resource or design choice changes communication and sensing. Their usefulness for another system depends on whether the relation survives the disturbances, hardware effects, and service demands of that system. Validation must therefore connect the controlled design choice to both measured outcomes and explain how the operating conditions were exercised. The central question is how much of that joint behavior has been demonstrated, from a numerical model to a deployed link, and how another group could test it again.

A. What Different Validation Settings Reveal

Figure 5 summarizes each study's highest reported setting. Laboratory and proof-of-concept work accounts for 78 studies,

and controlled prototypes for 66. Simulation or numerical work accounts for 32, enhanced simulation or dataset-based work for 18, and field trials or deployments for 12. These exclusive settings describe the reach of the reported evaluation; a single study can also combine analysis, simulation, and physical experiments. The complete overlapping method profile is retained in the electronic supplement.

Each setting exposes a different part of the coupling mechanism. Simulation can isolate waveform allocation and device sensitivities while holding other parameters fixed [73], [79]. Laboratory and prototype implementations expose conversion, synchronization, and receiver effects that enter the physical signal chain [62], [72], [115]. Field studies add the constraints of access networks, deployed fiber, submarine monitoring, and outdoor wireless links [22], [24], [33], [34], [116]. The scientific contribution of a field trial depends on which of these conditions it actually tests. A controlled prototype may sweep a larger range of a shared parameter or measure the two functions more directly than a field dataset collected for a single outcome.

The required disturbances follow the physical path. In fiber, topology, traffic, launch conditions, receiver operation, and the disturbance being sensed define the experiment. For VLC and LiFi, geometry, receiver orientation, illumination, and detector response determine the available observation. FSO adds pointing, target properties, atmosphere, and motion. Photonics-enabled terahertz systems cross an additional interface: optical conversion and reference stability must be related to antenna alignment and wireless propagation. Following the same configuration through analysis, simulation, and experiment helps distinguish a predicted resource interaction from an unmodeled component or environmental effect.

B. Field Coverage of Both Functions

Among the 12 field or deployment studies, six report outcomes in both functional domains at that level. They cover photonics-enabled terahertz and cellular links, fiber access integrated with radio-frequency identification, space laser ranging and communication, deployed multicore fiber, and submarine cable monitoring [22], [24], [33], [34], [48], [116]. Thus, field evidence for communication and sensing extends across several physical settings. This finding establishes coverage at the study level; it does not make all six systems equivalent joint demonstrations.

Concurrent joint operation requires the two outcomes to refer to the same configuration, time interval, and condition set. If a communication result comes from one test and a sensing result from another, their coexistence in a paper supports a different claim. Electronic Supplement S7 preserves the separate source locators and available timing information for the two functions. This distinction also matters when the environment is physical but the downstream processing is performed later. One fiber study combines field data with offline sensing classification [117]. The data show what was observed on the link, while the classification evaluates a subsequent processing stage; collection and functional execution occupy different parts of the evaluation.

TABLE V
REPORTED ARTIFACT ACCESS IN THE 206 STUDIES

Access route	Studies, <i>n</i> (%)
<i>Data</i>	
Closed or status unreported	145 (70.4%)
By request	41 (19.9%)
Open	13 (6.3%)
Outside artifact scope	7 (3.4%)
<i>Code or model</i>	
Closed or status unreported	197 (95.6%)
By request	7 (3.4%)
Partial components	1 (0.5%)
Outside artifact scope	1 (0.5%)

For a segmented system, the same reasoning extends across synchronization, conversion, and calibration interfaces. A joint test should show how each interface maps the transmitted signal and physical disturbance to the two reported outcomes. This makes it possible to identify whether a degradation originates in the shared optical resource, the wireless path, or subsequent processing, and to judge whether service continues during the disturbance.

C. Recovering the Experiment Behind a Result

Independent reconstruction requires more than a reported peak rate or error. Another group needs the waveform and processing chain, hardware configuration, scene and calibration, metric definitions, and data used to obtain that value. Table V summarizes the reported access routes. Closed access and an unspecified status are combined in the original coding and cannot be separated by interpreting this table.

Open data appear in simulation, laboratory, prototype, computer-vision, and field-data settings [79], [84], [115], [117], [118]. They permit reanalysis of the released observations. Repeating the physical experiment additionally requires the acquisition and calibration conditions, and reproducing a processed result requires the transformation from inputs to outputs. The computer-vision study reporting partial code or model components illustrates this distinction between an available component and a complete execution path [84].

We used the Technical Quality Assessment Framework (TQAF) developed for this review to describe such support. It scores eight technical and reporting dimensions from 0 to 3 and records overall evidence contribution separately. TQAF is a review-specific descriptive appraisal whose scoring has not been independently validated; it is not a conventional risk-of-bias assessment. The full profile and scoring rules are provided in the electronic supplement. Its reproducibility dimension combines parameter reporting with artifact access: 13 studies report open data, whereas three meet its strong reproducibility criteria. The difference reflects the additional information needed to reconstruct the system and its analysis.

D. From Reconstruction to a Shared Benchmark

A reconstructable experiment becomes a common test when independent designs can address the same task under a stated baseline, calibration, and metric definition. This requires

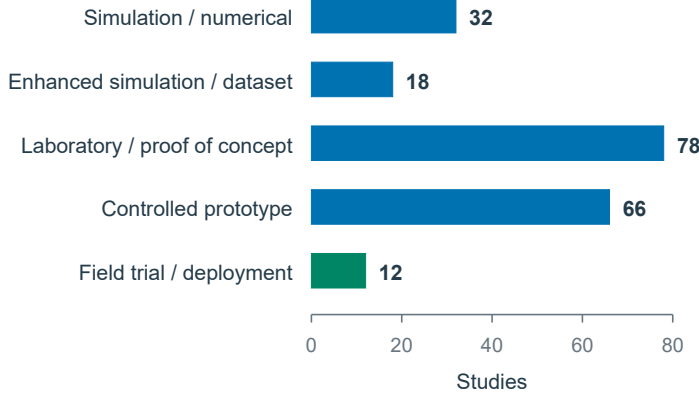
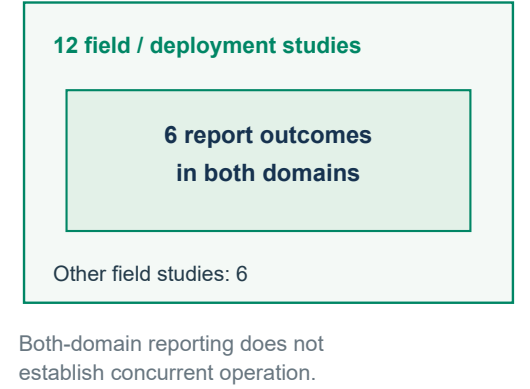
(a) Highest reported validation settingOne setting per study; $n = 206$ **(b) Field evidence, nested view**

Fig. 5. Validation reach in the 206 included studies. The highest-setting classes are exclusive. The nested field subset distinguishes 12 studies whose highest setting was field or deployment from six reporting field or deployment outcomes in both communication and sensing. Neither count establishes concurrent joint operation in one configuration and time interval.

agreement on which variables are controlled and which are allowed to change. For example, repeating a power-allocation result requires the same interpretation of the power budget; repeating a tracking result also requires the geometry and update process that produced the observation. The comparison conditions established in Section II are therefore part of the experiment.

No study reached the strong benchmark-readiness category under TQAF's combined criteria for common baselines, open artifacts, admissible outcomes in both domains, validation, and reproducibility. This review-specific result identifies an opportunity to combine capabilities already demonstrated separately. Shared tasks and recoverable experiments would allow the application requirements in Section VI to be tested against the same physical performance relation, rather than inferred from unrelated peak values.

VI. APPLICATION REQUIREMENTS AND NETWORK OPERATION

An application determines which part of the joint operating region is useful. A fiber monitor needs persistent observations while traffic continues; a moving terminal needs a position or channel estimate early enough to guide the next transmission. Both may reuse a signal, yet their limiting conditions differ. The mechanisms in Sections III and IV become system choices when these requirements are specified. This section follows that progression from the optical observation to a service outcome, using the validation distinctions in Section V to identify what each demonstration supports.

A. Continuous Monitoring Along a Communication Path

Fiber infrastructure provides a direct setting for sustained sharing: the medium carrying communication also responds to the environment being sensed. Phase, backscatter, and polarization can support distributed observations along a route.

Coherent multicore transmission with distributed sensing and operational submarine cable monitoring illustrate this reuse at different scales [24], [88]. Their application value depends on preserving a useful optical observation together with traffic continuity. Topology, launch and receiver conditions, and the disturbance context therefore belong to the service definition, rather than being incidental laboratory settings.

The sensing task changes the required observation and reference. A fiber platform for vehicles combines high-rate data transmission with distributed temperature measurement [119]. Infrastructure diagnosis can instead process collected fiber data into event classes [117]. Temperature estimates and event labels place different demands on calibration, observation duration, and reference data. An application evaluation should connect those demands to the communication interval affected by sensing. Offline classification can demonstrate information in a recorded signal, while timely operational diagnosis additionally depends on acquisition, processing, and delivery of the result.

Sharing extends beyond the sensing fiber when observations must be transported. One study relates transport capacity to the number of sensing streams meeting an echo-quality threshold [120]; another connects sensing traffic capacity with target count and resolution requirements [37]. These cases make a system-level coupling explicit: acquiring richer sensing observations creates traffic that consumes network resources. The useful operating point depends on the observations that reach the processing stage with sufficient quality, alongside the communication service maintained by the same infrastructure.

B. Mobility, Directional Links, and the Age of an Estimate

For a directional wireless link, sensing can inform where energy should be sent. A distributed W-band system uses radar estimates for beam alignment and flexible data service, while microcomb synchronization coordinates wireless data with radar imaging [121], [122]. These examples link spatial

information and a shared reference to communication operation. Their service interpretation depends on the complete interval from observation to transmission: the geometry at acquisition may differ from the geometry when a beam or resource decision is applied.

Observation age creates a pointing cost even when estimation error is small. Consider a sequential steering loop using an estimate referenced to the start of acquisition. Its age at actuation is

$$T_{\text{age}} = T_{\text{acq}} + T_{\text{proc}} + T_{\text{transport}} + T_{\text{act}}, \quad (6)$$

where the terms are acquisition, processing, transport, and actuation times. For constant relative transverse velocity $v_{\perp}$, nearly fixed distance d , and small angular displacement, uncompensated motion contributes

$$\Delta\theta_{\text{motion}} \simeq \frac{|v_{\perp}|T_{\text{age}}}{d}. \quad (7)$$

The remaining pointing budget must accommodate this displacement together with other errors at the selected service threshold. These illustrative kinematic relations explain why a precise but stale estimate can fail to support a narrow beam; they are not fitted results from the corpus. An overlapped pipeline uses the actual interval between the estimate's reference time and actuation. Prediction, acceleration, and measurement uncertainty require their own residual-error models.

The optical phased-array and frequency-comb demonstrations discussed earlier connect steering with ranging or imaging [51], [115]. In a moving application, narrower beams make acquisition, tracking, blockage, and calibration consequential for both functions. A useful evaluation therefore follows rate or reliability alongside sensing error and update timing as the geometry changes. A static alignment result identifies an achievable operating point; motion and recovery tests establish the conditions over which that point remains useful.

The steering mechanism must also remain physically explicit. The optical intelligent-surface study models a controlled mobile FSO path [100], whereas a laboratory demonstration uses mechanically rotated Risley prisms to steer a photonically generated terahertz signal [123]. Both redirect energy, but the control variables, propagation physics, and validation settings differ. Their future comparison requires a shared task and stated motion conditions, with actuator behavior and conversion interfaces retained in the timing budget.

C. Indoor Operation, Human Use, and Difficult Environments

In indoor optical systems, the user or robot's position and orientation shape both received power and the sensing geometry. Multicarrier positioning and robotic communication illustrate different consequences of this dependence [38], [101]. The robotics study reports packet loss over distance and view angle, display delay over queue size, and formation control under dynamic motion as separate results [38]. Together they show why communication reliability, information freshness, and control behavior need distinct measurements. They should not be collapsed into a single paired operating point unless their conditions and timing are linked by the source.

Human-facing optical devices connect communication or remote alerts with motion, falls, air pressure, physiology, or gesture [124], [125], [126]. Here, a sensing estimate is useful when it represents the intended event or physiological quantity under the user's motion and illumination conditions. Communication is useful when that information can be delivered with the required continuity. Reference measurements, event definitions, and processing delay consequently matter alongside link rate. This requirement also distinguishes a recognition result on collected data from an evaluated sensing-and-alert service.

Extreme environments impose a different set of physical constraints. Satellite-link work connects communication with atmospheric or wavefront sensing [52], making pointing and propagation part of the adaptation problem. Underwater optical-link concepts and subsea fiber monitoring involve different paths [112], [127]. Attenuation through water applies to the former; cable calibration and access to the installed infrastructure constrain the latter. Grouping both by their application environment is useful for locating requirements, but their physical observations and attainable operating regions remain distinct.

D. Inference and Control as Parts of the Joint System

Learning methods convert optical observations into positions, event labels, field estimates, channel predictions, or resource decisions. Fiber diagnosis, federated visible-light positioning, and wearable optical sensing illustrate this range [117], [126], [128]. The inference task adds a reference-data requirement to the physical measurement: reported accuracy depends on what was labeled, how data were divided, and the conditions represented in evaluation. Applying a learned model in another layout or operating regime therefore introduces a further test of the joint system.

One coherent-link model both compensates transmission impairments and estimates distributed power and abnormal loss [129]. The outputs share a link, but retain different targets and reference information. This architecture motivates testing whether a change in the link affects the two outputs similarly, and whether adaptation that helps one preserves the other. The answer cannot be inferred from communication compensation or sensing accuracy alone.

Processing also consumes time and resources. Classifier size changes recognition accuracy and the edge-processing burden in [130]. A digital-twin study represents vehicular edge resources and relates orthogonal task channels to application latency [131]. Another uses a model-aided physical reference for coherent fiber sensing [132]. These model roles differ: one organizes resources for a service, while the other supports interpretation of the physical observation. Reporting the represented state, update process, model mismatch, and resulting decision connects either role to a testable system benefit. It also allows inference accuracy to be weighed against the delay and resource use needed to obtain it.

E. From a Subsystem Benefit to a Network Role

Table VI connects these applications through the evidence needed to sustain a useful service. At network scale, an O-

TABLE VI

FROM APPLICATION OPERATION TO A JOINT DEMONSTRATION. THE OUTCOME PAIRS IDENTIFY MEASUREMENTS NEEDED TO EVALUATE THE STATED OPERATION; THEY DO NOT ASSERT THAT EVERY CITED STUDY MEASURED BOTH CONCURRENTLY.

Operation	Shared mechanism	Communication and sensing outcomes	Needed demonstration
Continuous fiber monitoring	Traffic and distributed observations use a common fiber or reference [24], [88]	Link continuity or BER; event, field, or disturbance estimate	Measure both during stated traffic and disturbance changes, preserving calibration and observation intervals.
Directional mobile service	Sensing informs steering; generation and reception share a reference [121], [122]	Delivered rate or reliability; range, angle, or motion estimate	Vary motion and alignment, including acquisition, update delay, blockage, and recovery.
Indoor positioning and control	Optical observations guide a location-dependent communication or control task [38], [101]	Packet delivery and delay; position or control error	Link the geometry and timing of each observation to the resulting data or control action.
Transport of sensing streams	Sensing quality and target coverage consume transport capacity [37], [120]	Carried traffic; delivered echo quality, target count, or resolution	Increase sensing and communication load together, identifying which service requirement first fails.
Learned sensing and adaptation	Processing converts a shared observation into estimates and decisions [129], [130]	Communication impairment or service measure; estimation or recognition quality	Apply a specified data or channel shift and record both outputs, inference cost, adaptation, and fallback.
Human use and difficult environments	Optical data and measurements share an interface with the user or environment [52], [112], [126]	Delivery continuity; task-specific estimate against a physical reference	Exercise the relevant motion, illumination, atmosphere, or cable condition with suitable reference data.

ISAC subsystem must occupy an explicit role in access, fronthaul, transport, edge processing, or service control. The consequences of sharing then include traffic, timing, orchestration, energy, and recovery across interfaces. For example, channel sensing can inform access placement under weather and demand changes [68]; evaluating its system benefit requires connecting the sensed condition to the placement decision and resulting communication service.

The corpus includes studies with an explicit 6G role [37], [115] and enabling contributions that require additional network context [28], [73]. The complete relevance classification and application inventory are provided in the electronic supplement. These classifications describe the literature's connection to 6G. Standard conformance, interoperability, and deployment readiness require evidence tied to the relevant network framework [6].

The remaining engineering question is whether the observed subsystem benefit persists after including the resources used to acquire, transport, process, and act on sensing information. Answering it requires the same two functional outcomes examined earlier, now measured under the load, motion, and timing of the intended role. Those unresolved links define the research tests developed in Section VII.

VII. RESEARCH QUESTIONS AND EVALUATION PRIORITIES

The preceding sections connect a physical sensing interaction to a shared design choice, two performance outcomes, and the conditions of an application. The unresolved issue is how far a useful relation persists when those conditions change. Greater integration can improve coordination, while also coupling distortion, synchronization, dynamic range, and failure behavior. Research can resolve this tension by identifying the mechanism that limits joint operation and testing the variables expected to move that limit.

A. From Reported Relations to Discriminating Experiments

The literature offers recurring mechanisms across optical platforms, but their quantitative consequences belong to particular tasks and measurement settings [11], [14], [15]. A useful next experiment should therefore begin with a relation already observed within a study. It should state which explanation is being tested, change the relevant parameter, and measure both functions against suitable separate-function baselines. The shared-resource budget must remain explicit: increasing a sensing probe with communication launch power held fixed answers a different question from redistributing a fixed total power budget.

Table VII identifies five research questions arising from the reviewed mechanisms and validation limits. The proposed tests distinguish an achievable operating point from the range of conditions over which that point remains useful. They are prospective experiments, not additional results of the review.

These experiments require perturbations that act on the native signal path. Fiber tests can vary topology, traffic, or the sensed disturbance; VLC tests can vary illumination and orientation; FSO tests can vary pointing and atmosphere. Photonic terahertz tests should separate optical drift from changes in wireless propagation before combining them. This separation helps explain whether the observed loss of joint performance follows from the proposed coupling mechanism or from a different component. Repeated paired trials then show the variability of the operating region under the stated conditions.

The result remains useful if integration provides no net gain. For example, if sensing-assisted alignment improves a static link but its update delay erases the advantage during motion, the crossover defines when the control method is appropriate. Likewise, a shared reference that improves one output while destabilizing the other identifies a need for compensation, a different allocation, or partial separation. Such findings turn

TABLE VII
TESTABLE RESEARCH QUESTIONS DERIVED FROM THE REVIEWED MECHANISMS AND VALIDATION LIMITS

Research question	Evidence motivating the test	Controlled test	Informative outcome
Which resource limit determines the joint gain?	Probe power and pilot allocation alter sensing support and communication impairment [1], [73].	Sweep probe fraction or pilot occupancy under declared power and bandwidth budgets; compare joint and separate-function baselines, measuring both outputs in each run.	The operating region and the interference, reference, or resource constraint that limits it.
How robust is a shared optical reference?	Coherent sensing and photonic dechirping depend on reference and receiver behavior [28], [61].	Introduce calibrated drift or timing mismatch, holding the task and processing definition fixed; repeat paired communication and sensing measurements.	The disturbance at which one function fails first and whether compensation restores both.
When does sensing-assisted control repay its overhead?	Radar-guided alignment and robotic queues connect observations to subsequent service [38], [121].	Vary motion, observation age, and control load; compare adaptive and reference operation with acquisition and update time included.	The conditions under which delivered service improves, and the delay or recovery boundary where the gain disappears.
Do learned joint functions survive a change of conditions?	Shared-link inference and classifier size connect estimation quality to processing cost [129], [130].	Apply a declared geometry, channel, or data shift; measure both outputs, inference delay, adaptation cost, and fallback against a fixed reference.	The range over which both functions remain useful and the resource cost of recovering from shift.
Can a relative design advantage be reproduced under service load?	Field coverage, artifact access, and transport capacity leave distinct validation gaps [37], [120].	Independently reconstruct one modality-specific task with common calibration and baselines; vary traffic and one physical disturbance during the same paired test.	Agreement or disagreement on the relative result, its limiting mechanism, and the load at which an application requirement fails.

a performance comparison into a design rule with an explicit range of validity.

B. Building Comparable Tests Across Groups and Networks

Independent comparisons need a common task within a physical modality, a reference design, calibration and ground truth, and sufficient artifacts to recover the reported processing. Agreement should concern the relative result and its conditions, with model bounds and empirical measurements retaining their different meanings. Such a test supplies evidence of transfer through an independently executed comparison.

At network scale, the experiment also needs the load, timing, and resource use of the intended role. Transporting observations, updating a model, or steering a beam can consume the resources that create the apparent subsystem benefit. Fielded cable and network work illustrates the importance of operational continuity [133], [134]. Future tests can connect that continuity to the sensing decision and include recovery from a component failure or corrupted observation. The resulting benchmarks would support discussion of network requirements while preserving the propagation and measurement conditions of each platform.

C. Scope and Limitations of the Review

The publication window, language requirements, and search sources in Section I-C delimit the literature represented here. Full texts were assessed for 272 of 330 sought reports, or 82.4%; findings from unavailable reports may be absent. Selection involved investigator judgment without routine independent duplicate assessment. The scientific methods supplement documents these limitations and the review conduct.

TQAF is a review-specific appraisal that has not been independently validated. Hybrid-system classification and thematic synthesis also involve judgment. Differences in tasks, metrics, measurement planes, and source protocols prevented

a common pooled effect model. No meta-regression, formal publication-bias assessment, or formal sensitivity analysis was performed. The synthesis therefore supports interpretation of reported mechanisms and conditions, rather than a universal effect, platform ranking, or estimate of worldwide technological maturity. These limits bound the proposed research priorities to the gaps identified in the retained literature.

VIII. CONCLUSION

O-ISAC design begins with the physical interaction that makes sensing possible and the resource that connects it to communication. A fiber can carry traffic and respond to a disturbance; an optical beam can transmit data and probe a scene; photonic hardware can create a shared reference for a wireless link. These roles lead to different constraints, even when the systems use similar waveforms or report similarly named metrics. Across the reviewed evidence, joint performance follows from how a design choice changes both the useful observation and its competing demands on power, bandwidth, timing, and receiver capability. Resource allocation can create interference or reduce payload capacity, while sensing information can also support transmission and control. The resulting benefit depends on the task, measurement plane, resource budget, and operating conditions. These mechanisms provide a basis for selecting and interpreting designs without implying a universal performance frontier across platforms.

The next step is to test whether these relations persist under the traffic, motion, propagation, and hardware conditions of the intended application. Paired measurements, suitable separate-function baselines, and reconstructable experiments can expose where a joint advantage holds, where it fails, and why. Extending those tests through acquisition, processing, and network control would connect optical integration to demonstrated service value.

REFERENCES

- [1] L. Maoqi, W. Jingchuan, L. Chen, Y. Changyuan, and L. Chao, "Pre-Compensation of LFM Sensing Probe via Integrated Sensing and Communication over Fiber System," in *2025 30th OptoElectronics and Communications Conference (OECC) and 2025 International Conference on Photonics in Switching and Computing (PSC)*, 2025, pp. 1–4.
- [2] Z. Li, J. Shi, C. Shen, Y. Zhang, J. Zhang, and N. Chi, "Adaptive visible light integrated sensing and communication for cable-free virtual reality," *Photonics Research*, vol. 13, no. 9, p. 2453, 2025.
- [3] J. Wang, L. Lu, J. Zhang, A. P. T. Lau, Z. Li, and C. Lu, "On the coexistence of distributed fiber optic sensing and IM/DD transmission via digital subcarrier multiplexing," *Optics Express*, vol. 34, no. 3, p. 5592, 2026.
- [4] S. A. H. Mohsan *et al.*, "Optical integrated sensing and communication: Fundamentals, applications, challenges and future aspects," *Optical Switching and Networking*, vol. 61, p. 100854, 2026.
- [5] D. Zhang *et al.*, "Integrated sensing and communications over the years: An evolution perspective," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 5014–5048, 2026.
- [6] International Telecommunication Union, "Framework and overall objectives of the future development of IMT for 2030 and beyond," ITU Radiocommunication Sector, Tech. Rep. Recommendation ITU-R M.2160-0, Nov. 2023.
- [7] Y. Chen *et al.*, "Optical Integrated Sensing and Communication Lidar Based on RBPSK-FMCW Joint Waveform and Optical Phased Array," *Journal of Lightwave Technology*, vol. 44, no. 10, pp. 4032–4040, 2026.
- [8] M. Zhou, R. Jin, S. Chen, J. Ding, M. Xu, and Z. Wu, "Turbulence-Resilient Communication through Beam Shaping Enabled by Multi-Region Turbulence Intensity Sensing," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025, pp. 1–5.
- [9] X. Guo, Z. Li, Z. Zhou, L. Deng, Y. Yu, and X. Zhang, "Millimeter-Wave Integrated Sensing and Communication System Based on a Photonics-Assisted Phased Array," *Laser & Photonics Reviews*, vol. 20, no. 10, p. e02350, 2026.
- [10] R. Kaide *et al.*, "Terahertz ISAC with Simultaneous Fast-Swept FMCW Radar and High-Speed Wireless Link Using a Single UTC-PD," *Electronics*, vol. 15, no. 8, p. 1608, 2026.
- [11] Y. Wen, F. Yang, J. Song, and Z. Han, "Optical integrated sensing and communication: Architectures, potentials and challenges," *IEEE Internet of Things Magazine*, vol. 7, no. 4, pp. 68–74, 2024.
- [12] C. Liang *et al.*, "Integrated sensing, lighting and communication based on visible light communication: A review," *Digital Signal Processing*, vol. 145, p. 104340, 2024.
- [13] Z. Lyu *et al.*, "Exploring photonic THz-ISAC systems with integrated waveforms," *Journal of Lightwave Technology*, vol. 44, no. 3, pp. 868–877, 2026.
- [14] X. Su *et al.*, "Optical integrated sensing and communication: From theory to practice," *Tsinghua Science and Technology*, 2025, available online 11 December 2025.
- [15] A. G. Khorasgani, "A survey on general and optical ISAC: Fundamental limits, design approaches, and future directions," *Optical Switching and Networking*, vol. 60, p. 100853, 2026.
- [16] Z. Liu *et al.*, "A review of optical fiber integrated sensing and communication systems," *Science China Information Sciences*, vol. 69, no. 6, p. 160302, 2026.
- [17] L. Lu, Y. Yan, B. Yang, Y. Yang, A. P. T. Lau, and C. Lu, "Towards converged networks: A review of integrated optical fiber sensing and communication," *Science China Information Sciences*, vol. 69, no. 6, p. 160301, 2026.
- [18] M. J. Page *et al.*, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. n71, 2021.
- [19] M. L. Rethlefsen *et al.*, "PRISMA-S: An extension to the PRISMA statement for reporting literature searches in systematic reviews," *Systematic Reviews*, vol. 10, no. 1, p. 39, 2021.
- [20] L. Wang *et al.*, "Integrated Communication and In-Band Spectrum Polarization-Based Sensing via Fraction-Division Non-Orthogonal Multiple Access," *Journal of Lightwave Technology*, vol. 44, no. 1, pp. 138–147, 2026.
- [21] J. Du *et al.*, "Flexible Waveform Design for RO-ISAC Based on OFDM-Embedded Maximum Length Sequence," *IEEE Wireless Communications Letters*, vol. 14, no. 10, pp. 3334–3338, 2025.
- [22] Q. Wang *et al.*, "Real-Time Integrated Sensing with 5.08 mm Range Resolution and 15.5 Gbps Bidirectional Wireless Communication Over a 270-Meter Outdoor Photonic-Assisted Terahertz Link," *IEEE Photonics Journal*, vol. 18, no. 4, p. 7301309, 2026.
- [23] X. Tang *et al.*, "Optical fiber communication and distributed multi-parameter sensing integrated network for smart power grid applications," *Science China Information Sciences*, vol. 69, no. 6, p. 160305, 2026.
- [24] S. Chen, K. Zhu, J. Han, Q. Sui, and Z. Li, "Photonic Integrated Sensing and Communication System Harnessing Submarine Fiber Optic Cables for Coastal Event Monitoring," *IEEE Communications Magazine*, vol. 60, no. 12, pp. 110–116, 2022.
- [25] H. Zhou *et al.*, "Ultrasonic phase extraction method for co-cable identification in coherent optical transmission systems," *Chinese Optics Letters*, 2024.
- [26] S. Gaiani, M. Fasano, A. Gatto, P. Parolari, P. Martelli, and P. Boffi, "Exploitation of FMF capabilities for joint communications and sensing," in *International Conference on Transparent Optical Networks*, 2025.
- [27] C. Guan *et al.*, "A novel passive bi-directional audio over fiber transmission system with 20 Km," *Optical Fiber Technology*, 2024.
- [28] H. He *et al.*, "Integrated sensing and communication in an optical fibre," *Light: Science & Applications*, vol. 12, no. 1, p. 25, 2023.
- [29] Z. Hu, C. Zhao, M. Zhang, J. Chen, and M. Tang, "Native and Reconfigurable Distributed Acoustic Sensing Integrated with Single-carrier 64QAM Transmission Signal," in *Optical Fiber Communication Conference (OFC 2025)*, 2025.
- [30] H. Zhu, X. Xu, Z. Liu, and J. Zhang, "Integrated optical covert sensing and communication," *Chinese Optics Letters*, 2025.
- [31] L. Gu *et al.*, "Co-wavelength-channel integration of ultra-low-frequency distributed acoustic sensing and high-capacity communication," *Photonics Research*, vol. 13, no. 6, p. 1611, 2025.
- [32] P. J. Urban, E. M. Miśkiewicz, D. Karczewicz, W. Kogut, and J. Kopczyński, "Parallel Path Effect-Free Remote Monitoring and Sensing in Live XGS-PON Using Legacy Optical Time Domain Reflectometry," *Journal of Lightwave Technology*, vol. 43, no. 14, pp. 6711–6719, 2025.
- [33] S. Zhang, J. Li, J. Zhu, D. Zhang, Y. Zeng, and Y. Luo, "First Demonstration of FTTR-based RFID Network for Integrated Sensing and Communication," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025, pp. 1–3.
- [34] Y. Zhao, J. Wan, S. Xu, and W. Zou, "Integration of endogenous sensing and communication based on photonics-enabled cellular base station networks with a spectrum fusion algorithm," *Optics Express*, vol. 32, no. 26, p. 47428, 2024.
- [35] A. Alzamil, M. Sharawy, E. Almohimmah, A. Ragheb, A. Almainan, and S. Alshebeili, "Development of an Integrated Communication and Sensing System Using Spread Spectrum and Photonics Technologies," *Photonics*, 2024.
- [36] M. Lei *et al.*, "A Spectrum-Efficient MoF Architecture for Joint Sensing and Communication in B5G Based on Polarization Interleaving and Polarization-Insensitive Filtering," *Journal of Lightwave Technology*, 2022.
- [37] A. Tzanakaki and M. Anastopoulos, "Optical Transport Networks Supporting Integrated Communications and Sensing in 6G," in *2025 European Conference on Optical Communications (ECOC)*, 2025, pp. 1–4.
- [38] S. Wang and H. Chen, "Optical Integrated Sensing and Communication for Cooperative Mobile Robotics: Design and Experiments," in *2023 21st International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt)*, 2023, pp. 207–214.
- [39] Y. Song *et al.*, "VehicleTalk: Lightweight V2V Network Enabled by Optical Wireless Communication and Sensing," in *2024 IEEE 99th Vehicular Technology Conference (VTC2024-Spring)*, 2024, pp. 1–5.
- [40] Z. Liu, J. Zhu, J. Yang, Y. Sun, Y. Yang, and J. Luo, "ReflexGest: Recognizing Hand Gestures Under VLC-Capable Lamps," *IEEE Transactions on Mobile Computing*, vol. 24, no. 7, pp. 6583–6594, 2025.
- [41] K. Dong, W. Cao, and M. Wang, "Optical Camera-Based Integrated Sensing and Communication for V2X Applications: Model and Optimization," *Sensors*, vol. 25, no. 22, p. 7061, 2025.
- [42] N. N. Luong, C. T. Nguyen, and T. V. Pham, "Performance Analysis of NOMA-Assisted Optical OFDM ISAC Systems with Clipping Distortion," in *2026 IEEE Wireless Communications and Networking Conference (WCNC)*, 2026.
- [43] R. Zhang, Y. Shao, M. Li, L. Lu, and Y. C. Eldar, "Optical Integrated Sensing and Communication With Light-Emitting Diode," *IEEE Internet of Things Journal*, 2025.
- [44] P. Dong, P. Tang, Y. Yin, L. Tian, and J. Zhang, "Analysis of Visible Light-Based ISAC Channel Characteristics in a V2V Scenario," in *IEEE Wireless Communications and Networking Conference, WCNC*, 2025.

- [45] B. Zhou, A. Liu, and H. So, "Visible Light Communication-Enabled Simultaneous Position and Orientation Detection for Harnessing Multipath Interference and Random Fading," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [46] Y. Cui *et al.*, "Retroreflective optical ISAC using OFDM: channel modeling and performance analysis," *Optics Letters*, vol. 49, no. 15, p. 4214, 2024.
- [47] L. Shi, Z. Liu, B. Béchadargue, H. Guan, L. Chassagne, and X. Zhang, "Experimental Demonstration of Integrated Optical Wireless Sensing and Communication," *Journal of Lightwave Technology*, vol. 42, no. 20, pp. 7070–7084, 2024.
- [48] A. Aboltins, V. Bespal'ko, and V. Kurtenoks, "Continuous Event Recording Technology For Integrated Sensing And Communications In Space," in *IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium*, 2024, pp. 2971–2974.
- [49] Z. Ye, L. Liu, Z. Lin, L. Liu, Z. Liu, and H. Fu, "Single-Source optical integrated sensing and communication (O-ISAC) with sequential target recognition," *Optics & Laser Technology*, vol. 198, p. 114961, 2026.
- [50] C. Carver *et al.*, "Catch Me If You Can: Laser Tethering with Highly Mobile Targets," in *Proceedings of the 21st USENIX Symposium on Networked Systems Design and Implementation, NSDI 2024*, 2024.
- [51] Y. Wen, F. Yang, J. Song, and Z. Han, "Joint Design of Optical Phased Array Beamforming and OFDM Waveform in Optical Wireless Integrated Sensing and Communication," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 9703–9718, 2026.
- [52] M. Wu, H. Yang, W. Liu, Q. Yao, J. Zhang, and M. Cheriet, "An Adaptive Control System for Satellite-Ground Laser Communication Based on Integrated Sensing and Communication," in *2025 30th OptoElectronics and Communications Conference (OECC) and 2025 International Conference on Photonics in Switching and Computing (PSC)*, 2025, pp. 1–4.
- [53] Z. Lyu *et al.*, "Multi-Channel Photonic THz-ISAC System Based on Integrated LFM-QAM Waveform," *Journal of Lightwave Technology*, vol. 42, no. 11, pp. 3981–3988, 2024.
- [54] J. He, S. Xiao, C. Ying, Z. Chen, and Y. Xiao, "Sliding Window-Based LSTM Scheme for PAM-4 Photonics-Assisted W-Band ISAC Systems," *IEEE Photonics Technology Letters*, 2026.
- [55] D. Shi *et al.*, "310Gbps Transmission and Centimeter-Level Ranging Photonic THz ISAC System," *Journal of Lightwave Technology*, 2026.
- [56] S. Ye *et al.*, "Photonic Frequency-Hopped THz ISAC-Enabled by Electro-Optically Tunable Laser," *IEEE Photonics Technology Letters*, 2026.
- [57] R. Kaide, S. Ye, Y. Wang, Y. Mikami, Y. Ueda, and K. Kato, "Integrated Terahertz FMCW Radar and FSK Communication Enabled by High-Speed Wavelength Tunable Lasers," *Photonics*, vol. 12, no. 10, p. 977, 2025.
- [58] R. Wang, W. Ma, J. Liu, H. Xu, Z. Shen, and W. Li, "Integrated sensing and communication system based on a photonic integrated broadband multi-beamformer," *Optics Express*, vol. 33, no. 10, p. 20619, 2025.
- [59] Y. Wang, W. Lu, Y. Song, and J. Yu, "Photonics-Assisted mmWave Integrated Sensing and Communication System Enabled by ODDM Waveform," *IEEE Transactions on Microwave Theory and Techniques*, 2026.
- [60] B. Dong *et al.*, "Photonic-Based W-Band Integrated Sensing and Communication System With Flexible Time-Frequency Division Multiplexed Waveforms for Fiber-Wireless Network," *Journal of Lightwave Technology*, 2024.
- [61] Z. Wang *et al.*, "Simultaneous Full-Photonic Dechirping and Demodulation THz-ISAC System Based on Thin-Film Lithium Niobate Modulator," *Journal of Lightwave Technology*, vol. 44, no. 11, pp. 4640–4647, 2026.
- [62] N. Tihomorskis *et al.*, "Software-Defined Radio Experimental Validation of an OTFS-Based ISAC for Velocity Estimation in an ARoF Setup," *Technologies*, vol. 14, no. 5, p. 262, 2026.
- [63] J. Wang, J. Zhang, A. P. T. Lau, and C. Lu, "Integrated ultra-large dynamic vibration sensing with fronthaul analog radio-over-fiber transmission," *Photonics Research*, vol. 13, no. 8, p. 2339, 2025.
- [64] A. Safiriyu *et al.*, "Heterogeneous Integrated III-V-on-SOI Transmitter for 6G FiWi mmWave/FSO Integrated Sensing and Communication," in *2025 European Conference on Optical Communications (ECOC)*, 2025, pp. 1–5.
- [65] H. Prabhakar, N. R. R., and V. A. Bohara, "A THz with VLC Assisted ISAC System Under the Impact of Human Blockages," in *2026 IEEE Wireless Communications and Networking Conference (WCNC)*, 2026, pp. 1–6.
- [66] R. Arunachalam, R. Singh, and M. V. Kumar, "Towards last-mile connectivity in 6G-IoT: An integrated MIMO-FSO communication system and FBG sensors under atmospheric attenuations, strain and temperature effects," *Engineering Science and Technology, an International Journal*, vol. 63, p. 101958, 2025.
- [67] M. Nafees, M. Baniyadi, J. R. Hopgood, M. Safari, and J. S. Thompson, "Integrated Sensing and Communication for UAV Trajectory Optimization in Mixed FSO-RF Networks in Dynamic Weather Conditions," in *2025 IEEE Wireless Communications and Networking Conference (WCNC)*, 2025, pp. 1–6.
- [68] —, "Leveraging ISAC for Adaptive Cell Zooming in Mixed FSO-RF Networks," in *2025 IEEE 36th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2025, pp. 1–6.
- [69] X. Song *et al.*, "Full dimensional control of structured microwaves based on microcombs," *Nature Photonics*, vol. 20, no. 4, pp. 377–385, 2026.
- [70] J. Shao, T. Z. Gutema, and W. O. Popoola, "Enhanced Room Coverage With Photon-Level NLOS Ultraviolet Optical Wireless Communication," *IEEE Photonics Technology Letters*, vol. 37, no. 24, pp. 1447–1450, 2025.
- [71] I. Krikidis, "Quantum Optical Integrated Sensing and Communication With Homodyne BPSK Detection," *IEEE Wireless Communications Letters*, vol. 15, pp. 1375–1379, 2026.
- [72] Y. Xia *et al.*, "High-Speed Flexible Near-Infrared Organic Photodetectors for Self-Powered Optical Integrated Sensing and Communications," *Advanced Functional Materials*, vol. 35, no. 2, p. 2412813, 2025.
- [73] M. F. Ahmed, R. U. Mondal, M. H. Kabir, and A. M. A. Rahman, "OFDM-based optical integrated sensing and communication framework using embedded pilot subcarriers," *Optics Continuum*, vol. 5, no. 6, p. 1706, 2026.
- [74] M. Che, B. Li, H. Tang, S. Ye, Y. Kamiura, and K. Kato, "Photonic Generation of Joint Amplitude-Frequency Modulated Waveform for THz Integrated Sensing and Communication," *Microwave and Optical Technology Letters*, vol. 67, no. 5, p. e70216, 2025.
- [75] M. A. Alloush and A. Sandmann, "Cc-Ofdr Sequence Shaping Enabling Joint Co-Directional Sensing and Communication," in *2025 IEEE Photonics Conference (IPC)*, 2025, pp. 1–2.
- [76] M. Lei *et al.*, "Photonics-aided integrated sensing and communications in mmW bands based on a DC-offset QPSK-encoded LFM CW," *Optics Express*, vol. 30, no. 24, p. 43088, 2022.
- [77] Y. Wen, F. Yang, J. Song, and Z. Han, "Optical Wireless Integrated Sensing and Communication Based on Optical Phased Array: Performance Metric and Optimal Beamforming," *IEEE Transactions on Wireless Communications*, vol. 24, no. 9, pp. 7221–7236, 2025.
- [78] Y. Zhu *et al.*, "Integrated 1-Tb/s/λ Coherent Optical Transmission and Forward Vibration Sensing Using DFB Lasers with Combined Linewidth of 21.4 MHz," in *2025 Asia Communications and Photonics Conference (ACP)*, 2025, pp. 1–5.
- [79] K. Deng, J. Liu, X. Lu, J. Chen, Y. Zhou, and W. Li, "A Unified OFDM-ISAC Signal Generation Architecture in W-Band via Photonics-Aided Frequency Multiplication and Phase Noise Mitigation," *Photonics*, vol. 12, no. 11, p. 1052, 2025.
- [80] G. Zhou, M. Cao, Q. Yang, Y. Zhang, S. Han, and H. Wang, "Integrated Design of OFDM-LFM Lidar Communication in IoV Environment," in *2024 IEEE 12th International Conference on Information, Communication and Networks (ICICN)*, 2024.
- [81] S. Chen *et al.*, "Microwave Photonics Empowered Minimalist Base Station for Integrated Sensing and Communication via Cascaded Intensity and Phase Modulation," *Journal of Lightwave Technology*, 2026.
- [82] B. Dong *et al.*, "Demonstration of photonics-based flexible integration of sensing and communication with adaptive waveforms for a W-band fiber-wireless integrated network," *Optics Express*, 2022, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
- [83] K. Matsuda *et al.*, "Sensing μm-scale vibrations in the Hz range within a THz communication system," *Optics Express*, 2026.
- [84] Z. Liu *et al.*, "CORE-Lens: Simultaneous Communication and Object REcognition with Disentangled-GAN Cameras," in *Proceedings of the 28th Annual International Conference on Mobile Computing And Networking*, 2022, pp. 172–185.
- [85] B. Yin *et al.*, "Integrated Sensing and Communication in Self-Homodyne MCF Transmission Systems With Linear Frequency Modulation," *Journal of Lightwave Technology*, 2026.
- [86] M. Lei *et al.*, "Photonics-assisted THz ISAC System Based on A Time-Frequency Efficient SF-LFM-OFDM Waveform," in *2026 Optical Fiber Communications Conference and Exhibition (OFC)*, 2026, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.

- [87] L. Wang *et al.*, "Endogenous integration of communication and interference fading free sensing using telecom pilots via joint polarization-fraction domain multiplexing," *Optics Express*, 2025.
- [88] X. Liu *et al.*, "Integrating High-Capacity Self-Homodyne Transmission and High-Sensitivity Dual-Pulse ϕ -OTDR with an EO Comb over a 7-Core Fiber," *Photonics*, vol. 13, no. 3, p. 261, 2026.
- [89] B. Yang *et al.*, "Accurate vibration recognition enabled by synergistic sensing based on both phase and SOP for stable IM-DD optical interconnects," *Journal of Optical Communications and Networking*, 2026.
- [90] M. Lei *et al.*, "Integration of Sensing and Communication in a W-Band Fiber-Wireless Link Enabled by Electromagnetic Polarization Multiplexing," *Journal of Lightwave Technology*, 2023.
- [91] T. Chu *et al.*, "Revolutionizing 6G: Experimental Validation of an Optical Integrated Communication, Sensing, and Power Transfer System," *IEEE Journal on Selected Areas in Communications*, vol. 44, pp. 689–701, 2026.
- [92] F. Liu *et al.*, "Millimeter-wave over fiber integrated sensing and communication system using self-coherent OFDM," *Optics Express*, 2024.
- [93] F. Liu, X. Zou, N. Zhong, X. Deng, L. Yan, and W. Pan, "Signal-Signal Beating Interference: From Destructive to Constructive for Photonic THz Integrated Sensing and Communication System Using Self-Coherent OFDM," *IEEE Transactions on Microwave Theory and Techniques*, 2026.
- [94] Y. Zuo, W. Liu, Y. Cheng, X. Li, and M. Luo, "High-performance Vibration Sensing Based on Subband Phase-Shift Transform in Integrated Sensing and Communication System," in *Asia Communications and Photonics Conference, ACP*, 2025.
- [95] J. Wang, A. P. T. Lau, and C. Lu, "Sensing as a Service: Can In-Band Distributed Fiber Optic Sensing Coexist With Coherent Metro DWDM Communication Network?" *Journal of Lightwave Technology*, 2026.
- [96] B. Dong *et al.*, "THz Integrated Sensing and Communication With Full-Photonic Direct LFM Reception and De-Chirping for D-Band Fiber-Wireless Network," *IEEE Transactions on Microwave Theory and Techniques*, 2025.
- [97] Q. Zhou *et al.*, "A Sub-THz ISAC System with Simultaneous Real-Time 125.516-Gbps Communication Rate and Offline 10-mm Sensing Resolution Enabled by Photonics," in *ECOC 2024; 50th European Conference on Optical Communication*, 2024.
- [98] X. Ou *et al.*, "Integrated Optical Sensing and Communication (IOSAC) System Based on Hybrid-Waveguide Structures," *Advanced Materials Technologies*, vol. 9, no. 3, p. 2300998, 2024.
- [99] L. Shi, B. Bechadergue, L. Chassagne, and H. Guan, "Joint Visible Light Sensing and Communication Using m-CAP Modulation," *IEEE Transactions on Broadcasting*, 2023.
- [100] H. Wang, Z. Zhang, Y. Ge, and B. Zhu, "Optical Integrated Sensing and Communication System Based on Combination of OIRS and PD Array for Mobile Scenarios," *IEEE Transactions on Wireless Communications*, vol. 23, no. 12, pp. 19 522–19 536, 2024.
- [101] H. A. Satai, B. Bechadergue, L. Chassagne, K. Acuna-Condori, B. Cagneau, and P. Ruau, "Single-Source Integrated Optical Wireless Positioning and Communication," in *2026 IEEE 6th International Symposium on Joint Communications & Sensing (JC&S)*, 2026, pp. 1–6.
- [102] X. Xiong, H. Wang, S. Shao, and C. Chen, "Hybrid MRR-Assisted Long-Range SDMA for RO-ISAC Systems in Industry IoT Applications," *Journal of Lightwave Technology*, 2026.
- [103] X.-X. Feng, M.-Y. Han, M.-P. Chen, Q. Fang, Y.-J. Wang, and X. Li, "Integrated Nitride optoelectronic chip for motion detection and visible light communication," *Chinese Optics*, 2023, eligible companion report in the governed 206-study/227-report O-ISAC evidence base.
- [104] Z. Xue, S. Li, X. Xue, X. Zheng, and B. Zhou, "Photonics-assisted joint radar and communication system based on an optoelectronic oscillator," *Optics Express*, vol. 29, no. 14, p. 22442, 2021.
- [105] Y. Hai, Y. Luo, C. Liu, and A. Dang, "Remote Phase-Shift LiDAR With Communication," *IEEE Transactions on Communications*, vol. 71, no. 2, pp. 1059–1070, 2023.
- [106] Z. Yue *et al.*, "Demonstration of Ranging-Based Phase Encryption for Enhanced Free-Space Optical Security," *Journal of Lightwave Technology*, 2026.
- [107] F. Yang, C. Lin, J. Gao, and Y. Chen, "Microwave Photonic Joint Radar, Wireless Communications, and Spectrum Sensing System With Broadband Tunability From 12 to 40 GHz," *IEEE Internet of Things Journal*, 2026.
- [108] Y. Wen, F. Yang, J. Song, and Z. Han, "Pulse Sequence Sensing and Pulse Position Modulation for Optical Integrated Sensing and Communication," *IEEE Communications Letters*, 2023.
- [109] W. Wang *et al.*, "Multiuser Retroreflective Optical ISAC: Waveform Design, Joint Sensing and Multiple Access," *Journal of Lightwave Technology*, 2026.
- [110] J. He, K. Liu, L. Yin, and Z. Chen, "An Orthogonal Time Frequency Space Waveform With W-MUSIC Scheme for Photonic-Aided ISAC System," *IEEE Photonics Technology Letters*, 2026.
- [111] J. He, L. Yin, K. Liu, S. Xiao, and Z. Chen, "Zero-Padding 2D-DFT Scheme for Photonics-Assisted W-Band Fiber-Wireless ISAC System With Orthogonal Time Frequency Space Modulation," *Journal of Lightwave Technology*, 2025.
- [112] Y. Wu *et al.*, "Advanced Integration of Sensing and Communication With High DSP Compatibility for SMART Network," *IEEE Photonics Technology Letters*, vol. 37, no. 21, pp. 1249–1252, 2025.
- [113] B. Yang, J. Tang, S. Gao, J. Liang, J. Wei, and Y. Yang, "Flexible Vibration Sensing Using Kalman Smoother Integrated in ECL-Based Communication Systems," *IEEE Photonics Technology Letters*, 2025.
- [114] G. Zhou, M. Cao, Q. Yang, Y. Zhang, X. Zhang, and H. Wang, "Bi-LSTM-Transformer hybrid network for V2V optical wireless ISAC systems," *ICT Express*, 2026.
- [115] M. Wang, W. Zhang, Z. Ren, S. Li, X. Zheng, and X. Xue, "Frequency-comb-steered ultrawideband quasi-true-time-delay beamformer for integrated sensing and communication," *Nature Communications*, vol. 16, no. 1, p. 7571, 2025.
- [116] Z. Wei *et al.*, "Simultaneous distributed sensing and communication enabled by space and wavelength division multiplexing in a field-deployed seven-core fiber," *Journal of Optical Communications and Networking*, vol. 18, no. 4, p. B49, 2026.
- [117] Z. Wang *et al.*, "Co-Route Fiber Recognition and Status Diagnosis Based on Integrated Sensing and Communication in 6G Transport Networks," *IEEE Internet of Things Journal*, vol. 11, no. 18, pp. 29 348–29 359, 2024.
- [118] T. Li *et al.*, "Submarine integrated sensing and communication cable networks enable high-precision earthquake location estimation through linear wavefront theory," *Journal of Optical Communications and Networking*, vol. 18, no. 4, p. B108, 2026.
- [119] F. Zhang *et al.*, "Glass fiber-based integrated sensing and communication system for vehicular applications," *Advanced Photonics Nexus*, 2026.
- [120] M. Anastasopoulos, J. Gutiérrez, and A. Tzanakaki, "Optical Transport Network Optimization Supporting Integrated Sensing and Communication Services," in *Optical Fiber Communication Conference in Proceedings Optical Fiber Communication Conference, OFC 2025 and Optical Fiber Communication Conference (OFC) Postdeadline Papers 2025*, 2025.
- [121] J. Jia, B. Dong, L. Tao, J. Shi, N. Chi, and J. Zhang, "Demonstration of radar-aided flexible communication in a photonics-based W-band distributed integrated sensing and communication system for 6G," *Chinese Optics Letters*, 2024.
- [122] X. Zhang *et al.*, "Microcomb-synchronized optoelectronics," *Nature Electronics*, 2025.
- [123] Z. Lyu *et al.*, "A Beam-Scannable Photonic THz-ISAC System Based on Risley Prisms," in *2023 Asia Communications and Photonics Conference/2023 International Photonics and Optoelectronics Meetings (ACP/POEM)*, 2023, pp. 1–4.
- [124] J. Wu, X. Lin, C. Yang, S. Yang, C. Liu, and Y. Cao, "Wearable Sensors Based on Miniaturized High-Performance Hybrid Nanogenerator for Medical Health Monitoring," *Biosensors*, 2024.
- [125] Z. Jiang *et al.*, "Optoelectronic integrated sensing and communication system based on InGaN/GaN MQW for wireless detection of air pressure," *Sensors and Actuators A: Physical*, vol. 395, p. 117086, 2025.
- [126] J. Li *et al.*, "An integrated sensing and communication solar skin for health monitoring and human-machine interaction," *Nano Energy*, vol. 148, p. 111639, 2026.
- [127] X. Li *et al.*, "Integrated Sensing and Communication Chip Based on III-Nitride for Motion Detection," *ACS Omega*, vol. 8, no. 16, pp. 14 656–14 661, 2023.
- [128] T. Wei, S. Liu, and X. Du, "Visible Light Integrated Positioning and Communication: A Multi-Task Federated Learning Framework," *IEEE Transactions on Mobile Computing*, vol. 22, no. 12, pp. 7086–7103, 2023.
- [129] F. Li *et al.*, "Double-effect DNN-based DBP Scheme for Integrated Sensing and Communications (ISAC)," in *2022 Optical Fiber Communications Conference and Exhibition (OFC)*, 2022.

- [130] J. Li *et al.*, “Indoor Health Monitoring with VLC-based Passive Posture Monitoring,” in *2024 Photonics & Electromagnetics Research Symposium (PIERS)*, 2024.
- [131] H.-N. Yang, J.-Y. Wang, Q. Zeng, M. Cheng, M. Lin, and J.-B. Wang, “Joint Task Offloading and Resource Allocation for Integrated VLC and Sensing in Digital Twin-Aided Vehicular Edge Computing Networks,” *IEEE Transactions on Vehicular Technology*, vol. 74, no. 4, pp. 6360–6372, 2025.
- [132] B. Yang, J. Tang, L. Fan, J. Wei, and Y. Yang, “Machine-learning-enhanced phase-based vibration sensing in coherent DSCM systems under commercial ECLs,” *Science China Information Sciences*, vol. 69, no. 6, p. 160306, 2026.
- [133] X. Cao *et al.*, “Intelligent submarine environmental monitoring based on fiber-optic integrated sensing and communication system,” *Science China Information Sciences*, vol. 69, no. 6, p. 160303, 2026.
- [134] E. Ip, Y.-K. Huang, M.-F. Huang, F. Yaman, J. Fang, and T. Wang, “Integrated sensing and communications over deployed terrestrial and submarine telecom cables,” *Science China Information Sciences*, vol. 69, no. 6, p. 160304, 2026.